THE ENTRUSTABLE PROFESSIONAL ACTIVITY (EPA) GUIDEBOOK FOR FACULTY AND RESIDENTS

Anaesthesiology EPA Guidebook, version 8. Last updated: July 26, 2024.

This public copy leaves out the original cover page, which listed the contributors and acknowledgements by name. The rest of the guidebook is unchanged.

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How to use this guidebook

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This guidebook is intended to inform both residents and faculty on Entrustable Professional Activities (EPAs) which will be implemented from July 2023.

All 3 anaesthesiology residency programmes have agreed upon the content of the EPAs and the minimum assessments that are stated under each EPA. The EPAs for anaesthesiology have also been approved by the Specialist Accreditation Board (SAB) as of 30 November 2022. Each residency programme however retains the right to variation in the implementation of the EPAs, for example by deciding if the levels of supervision are to be attained ahead of the timeline stipulated in the EPA or if the minimum number of Workplace-Based Assessments (WBAs) required to satisfy institutional accreditation is above the number stipulated in the EPAs.

Both residents and faculty are advised to familiarise themselves with the EPAs and the required assessments that come with it.

Since the start of residency, we have moved towards a Competency-based Medical Education (CBME) system that articulates the standard level of proficiency for all graduates, aligned with the roles graduates will need to fulfil in their careers.

Faculty will be familiar with WBAs such as the mini-Clinical Examination (mini-CEX) and Direct Observation of Procedural Skills (DOPS). For EPAs, WBAs have been modified to include entrustment questions so that the competency level of a trainee can be assessed not only from observation of the task or activity performed by the trainee, but also from the trainee's answers to questions that go beyond the scope of the task at hand i.e., entrustment questions.

The WBA forms that will be used in EPAs are found under the section on EPA assessment forms *(refer to page 7).*

The EPA assessment forms have different descriptors for the Likert scores from those used in the current WBA forms, that faculty may be familiar with. Both faculty and residents should read the section on EPA assessment forms for details.

Each component of the workplace-based assessment has also been mapped to sub-competencies stipulated in the milestones to aid the Clinical Competency Committee (CCC) in their assessment of the resident's progress and development.

You may find examples of entrustment questions that can be asked for each WBA under the entrustment questions section of each EPA. It is not necessary to ask all the entrustment questions suggested but at least 2 entrustment questions should be asked per WBA. The faculty has the prerogative to come up with his or her own entrustment question as long as it allows the faculty to discern and be able to decide on an appropriate level of supervision.

Workplace-Based Assessments are formative in nature and conducted so that the trainee can learn through and from the assessment. It is crucial that the EPA forms are filled soon after the assessment is completed, and feedback is provided. All forms should be filled within 24 hours of the assessment that is triggered by the resident to the faculty, failing which a reminder email will be sent to the faculty. A brief section on providing feedback provides basic principles and models that can be used by faculty as guides.

The words resident and trainee are used interchangeably.

Introduction to Entrustable Professional Activities (EPA)

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EPAs are defined as the core professional activities (work or tasks) that define a profession. For the Anaesthesiology specialty, 12 EPAs have been identified to be assessed during the residency training through Workplace-Based Assessments (WBA). Each EPA is a unit of professional practice (work or task) that may be entrusted to a learner to execute unsupervised once he or she has demonstrated the required competence.

Features of a trainee that enable trust include capability, integrity, reliability, humility, and agency. Capability encompasses the necessary knowledge, procedural and organisational skills, and clinical judgement to perform the professional task well. Integrity implies honesty and professionalism in keeping patient's welfare paramount even if it means admitting to errors or oversight and not withholding information. Reliable trainees are recognised for their sense of responsibility, conscientiousness, dependability in following through with tasks and the accuracy of information they provide. Humility is the ability to know one's limits in terms of knowledge and skills and be willing to ask for help in a timely and appropriate manner. It is also shown by the willingness to seek, receive and act on feedback. Agency refers to a trainee taking initiative and being proactive, whether in relation to patient care or personal professional development.

We recognise that EPAs cannot assess everything that a trainee must know to do, for example crisis management, and therefore serve as part of the programmatic assessment a trainee must complete.

To help faculty make an assessment and decision to trust, faculty need to **observe the resident's performance**, **discuss with the resident, and review what has been done by the resident** such as documentation in the electronic health records, case logs and reflection pieces.

For a resident to successfully complete the residency training and be recognised as a specialist, in addition to fulfilling the necessary clinical and cognitive experience and assessments such as the Master of Medicine in Anaesthesiology examinations and case log numbers, the stipulated level of entrustment must be attained for all 12 EPAs as shown in Table 1.

A description of the progressive levels of entrustments and the corresponding level of supervision is provided in Table 2.

Table 1: Expected Entrustment Levels by Residency Year

Anaesthesiology EPAs	EPA entrustment levels to attain in 5-year Anaesthesiology Residency	
	Level	**By when**
EPA 1: Performing preoperative/ pre-procedural assessment and optimisation of adult patients	Level 3	End R1
	Level 4	End R2
EPA 2: Providing anaesthetic care for adult patients requiring general anaesthesia	Level 3	End R2
	Level 4	End R3
EPA 3: Providing anaesthetic care for patients under regional anaesthesia (RA)	**For patients under RA (including single shot peripheral nerve blocks and subarachnoid blocks)**	
	Level 3	End R2
	Level 4	End R3
	For patients under RA with catheter	
	Level 3	End R3
	Level 4	End R5
EPA 4: Assessing and managing patients with potentially difficult airway in the hospital setting	Level 3	End R3
	Level 4	End R5
EPA 5: Assessing and managing patients with polytrauma in the perioperative setting	Level 4	End R3
EPA 6: Managing critically ill patients in the intensive care unit (ICU)	Level 3	End R2
	Level 4	End R3
EPA 7: Providing anaesthetic care for obstetric patients	Level 3	End R3
	Level 4	End R5
EPA 7a: Managing parturient for labour epidural/ combined spinal epidural (CSE)	Level 3	End R3
	Level 4	End R5
EPA 7b: Managing parturient for Lower Segment Caesarean Section (LSCS)	Level 3	End R3
	Level 4	End R5
EPA 8: Providing anaesthetic care for patients in a non-operating theatre setting	Level 3	End R3
	Level 4	End R5
EPA 9: Providing anaesthetic care for paediatric patients aged between 3 to 16 years old	Level 3	End R3
	Level 4	End R5
EPA 10: Managing adult patients for cardiovascular procedures and interventions.	Level 3	End R3
EPA 11: Assessing and managing patients with acute pain	Level 3	End R3
	Level 4	End R5
EPA 12: Assessing and managing patients with chronic pain	Level 3	End R5

Table 2: Description of levels of entrustment and corresponding levels of supervision

Level 1	Observe only	The trainee has permission only to be present but not to enact this EPA.
Level 2	Direct supervision	Supervisor needs to be present in the SAME room as the resident at all times. Pro-active supervision is required for all / most clinical activities.
Level 3	Indirect supervision	Supervisor need not be in the same room as resident but remains within the healthcare facility, contactable and ready to respond within minutes to any call for help by the resident e.g., the supervisor having a meal break while the resident remains alone to care for the patient
Level 4	Distant supervision	Supervisor is not present in healthcare facility but remains contactable via phone call to provide advice and supervision remotely.
Level 5	Supervise	The trainee is able to guide others in this EPA.

Residents will exit the residency training at level 4 entrustment in all EPAs with the exception of EPA 10 and 12 at level 3 entrustment because these require additional fellowship training to attain level 4 entrustment.

The structure of an EPA

Each EPA has

1. A title that states the EPA.
2. Specifications and limitations that describe the EPA and qualify what is included in this EPA.
3. Potential risk in case of failure which states the risks to patient and healthcare system if the EPA is practised poorly.
4. Most relevant domains of sub-competency that describe patient care, medical knowledge, interpersonal communication skills, practice-based learning, professionalism, or systems-based practice competencies required of the trainee for successful practice of the EPA.
5. Knowledge, skills, attitudes, and experiences section that states stipulated specifics of these required for each EPA.
6. Sources of information which include case log numbers, written reflections, assessments and course/ workshop certification or attendance, provide documented evidence of experiential learning in the EPA.
7. Level of entrustment that stipulates the timeline for attainment of the expected level of entrustment and supervision. This allows the CCC to track the progression of the residents and allow for early interventions to ensure the resident has the necessary exposure, experience, and knowledge in order to attain the expected level of entrustment by the expected timeline. If the trainees do not meet the expectations, it is likely that CCC will not sanction the progression of the trainee across residency years.
8. Expiration date for the EPA is standardised at 1 year across all EPAs for Anaesthesiology. This means that if the trainee is absent from clinical training for more than a year, he/she will assume a lower level of entrustment and supervision upon resumption of clinical training. The trainee will be able to resume the EPA at a higher level of entrustment and supervision after completing the necessary assessments.

A full description of the EPA serves as a guide for both trainee and supervisors to know what the individual is ready and permitted to do within the healthcare system.

EPA Assessment Forms

If there is any conflict of interest with regards to assessing the resident, faculty should highlight to the residency programme. Residents should also not choose assessors who may be conflicted in their assessment of the resident e.g., close relations.

The EPA assessment forms used are:

a. Mini-Clinical Examination (Mini-CEX) Form 

b. Direct Observation of Procedural Skills (DOPS) Form 

c. Entrustment-based Discussion (EBD) Form 

d. Competency Evaluation Form 

e. Multisource 360° Feedback (MSF) Form 

The EPA assessment forms will be administered via MedHub. All faculty and residents should have a MedHub account which will require a 2-point verification mode that will be linked to your work email. It is advisable to download the MedHub mobile app for convenience. Due to corporate firewalls, for the initial encounter with Medhub, you may find Medhub under spam or quarantine. To facilitate Medhub from being blocked in the future, at the quarantine email click on release and allow sender. Please note that the laptop version of Medhub will defer from the mobile version. For more information on how to use MedHub via laptop or phone, please refer to the attachment. 

The resident is expected to take ownership and ensure that all relevant mini-CEX, DOPS, and EBD for an EPA are completed by the end of each clinical rotation. Programmes will provide residents with instructions for MSF 360 evaluation forms. Programmes will also match faculty for the competency evaluation form. Only physician faulty/ core faculty may be assessors for mini-CEX, DOPS and EBD. Senior residents will not be involved with the assessment of EPAs.

During the WBA, the assessor should decide if the resident has been observed to:

1. Be proactive in reflective learning and self- improvement, receptive to feedback.
2. Be honest, reliable, conscientious, responsible.
3. Recognise own limits and escalate to supervisor when appropriate.

4. Provide efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality.
5. Be a collegial and respectful team player who is willing to help others.

Mini-CEX/ DOPS/ EBD forms

These forms are specific to the EPA and are available for selection in MedHub. Both faculty and resident must agree on the case for assessment. After the observed WBA, the resident must trigger the EPA form by matching it to the assessor. The case specifics should be filled in by the resident before the faculty fills up the form. The same patient can be used for at most 2 EPA assessments as agreed with the faculty, provided all the relevant specific forms are filled up separately e.g., the same patient with respiratory disease can be used for EPA 1 mini-CEX for preoperative assessment of a patient in the hospital setting e.g. preoperative clinic/ pre-anaesthetic evaluation clinic (PEC), and EBD for respiratory disease, as long as both the mini-CEX form for preoperative assessment form and the EBD form for assessment of a patient with pre-existing respiratory disease are completed.

For mini-CEX and DOPS, the assessment is done by faculty observing the resident's performance with minimal interference unless the patient's well-being and safety are at risk.

The scoring system for the observation portion of the mini-CEX and DOPS are the same. These are based on a 9-point Likert scale with the following descriptors:

1. Trainee cannot perform this activity.
2. The trainee can perform this activity only under direct supervision with supervisor performing more than 50% of activity.
3. The trainee can perform this activity only under direct supervision with supervisor performing less than 50% of activity.
4. The trainee can perform this activity but requires occasional direct supervision.
5. The trainee can perform this activity under indirect supervision for uncomplicated case.
6. The trainee can perform this activity under indirect supervision for complicated case.
7. The trainee can perform this activity independently with distant supervision.
8. The trainee performs this activity independently with distant supervision well enough to supervise others.
9. The trainee performs this activity independently with distant supervision well enough to instruct others.

This scoring system is different from the current Likert scale that is based on the faculty's expectation of the resident's performance as compared to his/ her cohort of the same year. In this new system, for each subcomponent activity of the mini-CEX e.g., patient assessment/ preparation or preparation of physical location, the faculty is expected to assess the level of supervision required based on the observed performance.

If a subcomponent activity is not observed, please choose the option of not observed.

For the trainee to attain level 3 entrustment, a minimum score of 5 for overall clinical care or assessment is required for indirect supervision. For DOPS, this would correlate with satisfactory or meeting expectation.

For the trainee to attain level 4 entrustment, a minimum score of 7 for overall clinical care or assessment is required that would correlate with distant supervision.

For EBD, the assessment is done by faculty who has co-managed the case with the resident. Based on observation of the resident, there may be areas for improvement with regards to patient assessment and preparation, preparation of the physical location, clinical management plan, communication skills with patient/ next of kin, communication skills with healthcare team, technical skills, clinical judgment, organisation, and efficiency or professionalism. If there are areas for improvement, please select the appropriate choice and fill the comment section with specifics that will be useful for the resident. If there are no obvious areas for improvement, please select this choice before proceeding further.

For the entrustment-based discussion section for mini-CEX and EBD, the descriptors for the 9-point scale are different from the observation portion. Medical knowledge, clinical reasoning and where possible, practice-based learning and systems-based practice competencies are assessed based on milestone descriptors. It is compulsory for medical knowledge and clinical reasoning to be filled. A minimum of 2 entrustment questions should be asked and faculty must describe briefly what was discussed.

Faculty are expected to record the supervision level that the faculty would **theoretically** entrust the resident with for this particular EPA.

Level 1	Observe only	The trainee has permission only to be present but not to enact this EPA.
Level 2	Direct supervision	Supervisor needs to be present in the SAME room as the resident at all times. Pro-active supervision is required for all / most clinical activities.
Level 3*	Indirect supervision	Supervisor need not be in the same room as resident but remains within the healthcare facility, contactable and ready to respond within minutes to any call for help by the resident e.g., the supervisor having a meal break while the resident remains alone to care for the patient
Level 4*	Distant supervision	Supervisor is not present in healthcare facility but remains contactable via phone call to provide advice and supervision remotely.
Level 5	Supervise	The trainee is able to guide others in this EPA.

**Residents will exit the residency training at level 4 entrustment in all EPAs with the exception of EPA 10 and 12. Resident will exit EPA 10 and 12 at level 3 entrustment because these require additional fellowship training to attain level 4 entrustment*

During the feedback, please comment on what the resident did well in and what can be improved upon, including resident's readiness to handle variations of the situation or clinical case described above**. Please also document the resident's discussion with faculty and reflections on the case.**

Faculty will be requested to indicate how receptive the resident was to feedback and if the resident demonstrated reflective learning via the dropdown box before submitting the form. A robust discussion and display of critical thinking skills should be encouraged.

Guide for DOPS

Please refer to the attached document below for details on the expectations for DOPs, what is considered below expectations, meeting expectations and or exceeding expectations based on what is observed.



Guideline &
Expectation for DOF

Multisource 360° Feedback (MSF) Evaluation Form

The resident will be instructed by programme if they are expected to conduct the matching of assessors for a particular set of MSF evaluation. Please note the number to be completed satisfactorily under "Sources of information to support summative entrustment decisions" section refers to a set of MSF evaluations for a stipulated rotation or time period as defined by the programme or EPA.

For example, 3 in ICU setting for EPA 6 refers to 2 sets of MSF evaluations to be completed during Junior Residency and 1 set of MSF evaluation to be completed during Senior Residency. Each set of evaluation will comprise of minimally 1 faculty, 1 peer and 2 nurses for EPA 6 but the composition of assessors may differ for each EPA. If the composition of assessors is indicated within the EPA document, it should be adhered to.

The 5-point scale requires the assessor to rate the resident according to how well the stated characteristics describe the doctor's behaviour in each section.

Competency Evaluation Form

The competency evaluation form follows the same 5-point scale for MSF form. Narratives offer a richer evaluation of the resident and are strongly encouraged. There is the option to indicate confidential comment to the PD if there are serious concerns about the resident.

Abbreviations for title of EPA assessment on MedHub

Each EPA assessment is specific and may be selected on Medhub based on the assessment type. Within the title of the EPA assessment, there is a description of the assessment scope and the expected level of entrustment for the assessment. For example, [EBD] EPA 1 (Level 3): Assessment of a patient with pre-existing respiratory disease. Due to the 100-character limit for titles of EPA assessment in the system, abbreviations are used in the title of the assessment.

For the full description of the EPA assessment, please look under the assessment section of each EPA for details on the assessor and the scope of the assessment.

Mx:	Management
Em:	Emergency
PNB:	Peripheral Nerve Block
CSE:	Combined Spinal Epidural (this will also include epidural)
CPB:	Cardiopulmonary bypass
Spinal:	Subarachnoid block
LSCS:	Lower segment caesarean section
OLV:	One Lung Ventilation
Req:	Requiring
GA:	General Anaesthesia
MRI:	Magnetic resonance imaging
Level x:	Entrustment level x
R1:	end of R1
R2:	end of R2
R3:	end of R3
R4:	end of R4
R5:	end of R5

Entrustment Questions

Entrustment questions may come at the end of a direct observations assessment such as a mini-clinical examination (mini-CEX), direct observation of procedural skills (DOPS) or during an entrustment-based discussion (EBD) of a case that is co-managed by the faculty doing the assessment. They aim to allow the assessor to discern and be able to decide on an appropriate level of supervision for the resident based on clinical reasoning and knowledge, which may extend to systems-based practice or evidence-based practice.

It is important to note that these discussions need to follow an actual case.

The 4 main questions to ask and be answered by the end of the assessment are:

1. What did the learner do/ does the learner know what to do?
    a. This usually involves getting the learner to describe **what** he/she has done to demonstrate knowing the steps or process in managing the case.
2. Does the learner have adequate background knowledge?
    a. This usually involves getting the learner to describe **why** he/she does the various steps to demonstrate knowing the reasons for the various steps in managing this case. This should normally be done concurrently with demonstrating what to do.
3. Is the learner aware of the risks and possible complications of the activity i.e., **what could go wrong**? Is he/she able to display adaptability to these different scenarios?
    a. The assessor may ask "What are some of the common or known problems, risks or complications that can happen to this patient undergoing this procedure?"
    b. The assessor may ask “How will you handle the situation if you are managing this case by yourself?”
4. What would the learner do in case of **unusual** patients, rare findings, emerging risks, or complications?
    a. These are the “What ifs” type of questions. Example of “what ifs” questions are:
        i. If the patient has a different medical problem or physiological feature such as obesity, pregnancy, poorly controlled DM, recent URTI or COVID infection etc.
        ii. If there is an unexpected surgical or laboratory findings
        iii. If the case presents at night

A minimum of 2 entrustment questions should be asked for each WBA. Faculty must describe briefly what was discussed.

Please read the specifics and the discussion points for each assessment so that even if the suggested entrustment questions are not used, the entrustment questions asked by faculty will not overlap in scope with the other EBD or WBAs with entrustment questions for the EPA.

Examples of entrustment questions can be found under the respective EPA in the section titled Entrustment Questions for EPA x.

Examples of Generic Entrustment questions for Evidence-based Medicine (The bracketed numbers suggest questions relevant to the milestone scoring)

1. Where would you be able to locate evidence to guide you decision making for this patient? (1)
2. How would you integrate evidence-based medicine or guidelines with patient preference to aid you in the management of this patient? (5)
3. Do you know of any studies that could guide you in the management of this patient e.g., in blood management decisions? (3)
4. If this patient had this condition or stipulation that will affect management, how would you use current evidence to optimise management while respecting the values and preference of the patient e.g., Jehovah's witness undergoing major surgery with potential massive blood loss, how would you use current evidence to plan your fluid management (5)
5. If current literature suggest opposite effects of an intervention, how would you decide what to do for this particular patient? (7)

Examples of Generic entrustment questions for Systems based Practice (The bracketed numbers suggest questions relevant to the milestone scoring)

1. How was care coordinated between the different disciplines for this patient? (7)
2. Who are the stakeholders and how can we ensure that all stakeholders are on the same page with regards to care of this patient? (3)
3. How would our anaesthetic plan affect the surgical plan or patient recovery? (5)
4. How can we ensure that the patient receives the best care without falling through cracks in the system? (7)
5. How do you think handover of care of the patient should be done? (7)
6. How would you ensure that the patient remains pain free after being discharged from the hospital back home? Is there any way that pain can be managed better in the community setting? (7)
7. What is value driven care and how is it applied to this patient? (7)

Providing feedback

Feedback is an essential part of assessment. Without provision of constructive feedback, learning may not result from the assessment.

The principles of providing feedback include:

1. Timeliness. Provide feedback soon after the observation.
2. Focus on observed actions and behaviours rather than directed at the individual's character, personality, or values.
3. Encourage self-assessment first e.g., "how do you think that went."
4. Avoid facial expressions and body language that can give generalised feedback in a non-verbal way.
5. Be clear and specific.
6. Include positive feedback on things done well together with constructive feedback on things that can be done better to create a positive and supportive learning environment; positive feedback is important to reinforce desirable actions or behaviour.
7. Limit the feedback to the amount of information that the trainee can take in; focus on 2 or 3 points at most to work on.
8. Encourage reflection on your feedback by asking what all this means to the resident
9. Choose to use "I" rather than "you" in your feedback e.g., I wonder, I observed, I thought, as "you" can carry a judgemental connotation and potentially reduce acceptance of the feedback.
10. Practice mutual respect and listening to each other
11. Where feasible, develop a learning plan together or ask what the resident will do differently next time
12. Consider the use of feedback models.

Feedback models

1. **DESC**

   Describe objectively what action or behaviour you observed which requires a change

   Express how you perceived the action affected others

   Suggest how to improve the situation

   Outline positive **C**onsequence (s) of improvement

2. **ADAPT**

   Ask

   - Ask learner to assess own performance
   - Assess learner's insight
   - Tailor feedback to learner perceptions
   - Be an active listener: ask questions and reflect back

   Discuss

   - Learner's self-assessment
   - Your observations about the performance: descriptive, specific, positive, corrective

   Ask

   - Ask what feedback means for the learner
   - Ask about opportunities for development and change

   Plan **T**ogether

   - Ask for the action plan
   - How can the learner get from point A to point B
   - How can you help the learner

Logbook

The training portfolio includes case records which encompass both case reflections and case log numbers. These will be reviewed at regular intervals by the program. For the EPAs that require case numbers and case reflections as part of the assessment, the minimum numbers will have to be fulfilled by the stipulated timeline before the level of entrustment is determined.

Further details on how to do a good case reflection and how to log case numbers will be provided in the training portfolio.

Individual EPA Templates

Anaesthesiology EPA 1

<table>
<tr><td>Title</td><td>Performing preoperative/ pre-procedural assessment and optimisation of adult patients</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td>Specifications:
This EPA includes both elective and emergency procedures/ surgery.
1. Establishing rapport with patients
2. Gathering data on disease processes and medical issues relevant to anaesthetic care
3. Identifying the need for, ordering, and interpreting appropriate pre-procedural investigations and following up of results
4. Performing anaesthetic risk assessment and determining fitness for surgery
5. Optimising patients preoperatively with inter-professional teams
6. Referring patients to other specialties for further evaluation and optimisation as necessary
7. Counselling and obtaining appropriate informed consent for anaesthetic care
8. Counselling patients who are breastfeeding regarding anaesthesia and analgesia options
9. Organising and documenting all relevant information to facilitate effective handover
10. Proposing an appropriate anaesthetic management plan based on preoperative assessment</td></tr>
<tr><td>Limitations:
This EPA excludes the following:
1. Patients less than 16 years’ old
2. Obstetric patients</td></tr>
<tr><td>Sub competencies relevant for this EPA</td><td>
<table>
<tr><td></td><td>PC1</td><td>✓</td><td></td><td>PC7</td><td></td><td></td><td>SBP1</td><td></td><td></td><td>P1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC2</td><td></td><td></td><td>PC8</td><td></td><td></td><td>SBP2</td><td>✓</td><td></td><td>P2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC3</td><td></td><td></td><td>PC9</td><td></td><td></td><td>SBP3</td><td>✓</td><td></td><td>P3</td><td></td><td></td></tr>
<tr><td></td><td>PC4</td><td></td><td></td><td>PC10</td><td></td><td></td><td>PBL1</td><td>✓</td><td></td><td>ICS1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC5</td><td></td><td></td><td>MK1</td><td>✓</td><td></td><td>PBL2</td><td>✓</td><td></td><td>ICS2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC6</td><td></td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td></td><td></td><td>ICS3</td><td>✓</td><td></td></tr>
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Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	1. Basic medical conditions, related complications and implications for anaesthesia including: • Cardiovascular i. Hypertension and ischaemic heart disease ii. Valvular heart disease iii. Rhythm disorders and conduction defects iv. Cardiac implantable electronic devices v. Cardiac failure and cardiomyopathy vi. Cardiac tamponade and pericardial effusion vii. Congenital heart disease viii. Pulmonary embolism • Pulmonary 1. Obstructive disease 2. Restrictive disease 3. Pulmonary hypertension 4. Infection e.g., upper and lower respiratory tract infections 5. Tobacco usage 6. Obstructive sleep apnoea • Neurology/ Muscular i. Cerebrovascular accidents ii. Epilepsy iii. Parkinson's disease/ cerebellar disease iv. Dementia v. Spinal cord injuries vi. Peripheral neurological disorders vii. Primary muscle disorders • Renal i. Renal failure • Liver i. Liver failure • Endocrine i. Diabetes mellitus ii. Thyroid disease iii. Pituitary disease iv. Adrenal disease v. Phaeochromocytoma vi. Carcinoid syndrome vii. Chronic steroid use • Haematology i. Anaemia ii. Polycythaemias iii. Clotting disorders iv. Antiplatelets and anticoagulants • Cancer i. Disease, therapies, and toxicities • Anaesthesia related genetic conditions i. Pseudocholinesterase deficiencies ii. Malignant hyperthermia • Others i. Obesity ii. Immunocompromised patients iii. Post organ transplant iv. Geriatric patients 2. Risk stratification • Impact of medicines and drug interactions in the perioperative period • Factors improving surgical and long-term outcomes • Risk stratification and scoring tools to stratify surgical and anaesthetic risks • Common risk assessment tools for various diseases and complications

		3. Risk mitigation • Risk mitigation strategies and appropriate interventions • Optimisation strategies for smokers • Role of prehabilitation and patient optimisation • Role of nutrition in the perioperative period • Principles of enhanced recovery pathways • Strategies to minimise use of blood products
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Ability to navigate electronic medical records and access best practice guidelines to guide optimisation of patient. 3. Communication and interpersonal skills: • Patient-centric communication including counselling and taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team huddling and accurate documentation 4. Non-technical skills (e.g., ability to organise, time management, resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides patient-centric compassionate care, ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Perioperative assessment of patients as part of operating theatre/ core anaesthesia training rotations – both in the emergency and elective setting 2. Pre anaesthesia clinic rotation (minimum 2 weeks) 3. Teaching session including simulated patient encounters
Potential risks in case of failure	1. Inadequate patient preparation for surgery/ procedures resulting in postponement or cancellation of procedure 2. Increase in patient morbidity/ dissatisfaction/ mortality 3. Increase in medico-legal liability 4. Increase in healthcare costs	

Anaesthesiology EPA 1 *(Click here to return to Table of Content)*

	Tools	Number to be completed satisfactorily	Additional specifications if needed (To consider evaluators and context)
Sources of information to support summative entrustment decisions	Mini-Clinical Examination (Mini-CEX)	Minimum 1 by end R1 (For level 3 supervision)	Physician Faculty *To be completed by end R1* 1. Preoperative assessment of a patient in the hospital setting e.g., preoperative clinic/ pre-anaesthetic evaluation clinic (PEC), ward
	Entrustment-based Discussion (EBD)	Minimum 3 by end R1 (For level 3 supervision)	Physician Faculty co-managing case *To be completed by end R1* 1. Assessment of a patient with pre-existing respiratory disease 2. Assessment of a patient with pre-existing cardiovascular disease 3. Assessment of a patient with pre-existing endocrine disease
		Minimum 2 by end R2 (For level 4 supervision)	*To be completed by end R2* 1. Assessment of geriatric patient 2. Assessment of morbidly obese patient
	Multisource 360° Feedback (MSF)	1 during pre-anaesthetic clinic rotation	Minimum 2 nurses and 2 doctors
	Competency Evaluation	Minimum 2 per year	Physician Faculty
Which supervision level when?	Level 3 supervision by end R1 Level 4 supervision by end R2		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 1

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Preoperative assessment of a patient in the hospital setting e.g., preoperative clinic/ pre-anaesthetic evaluation clinic (PEC), ward	1. Assessment of post-operative nausea and vomiting risk and strategies for mitigation 2. Ability to differentiate between drug allergy vs side effect 3. Evaluation of a patient with symptoms of an upper respiratory tract infection a. Differentiate between upper respiratory tract infection and allergic rhinitis b. Knowledge of potential complications c. Rationale for cancellation/ decision to proceed 4. Transfer of relevant information within the healthcare system e.g., dental work, need for further evaluation/ investigations 5. Counselling for anaesthetic techniques (knowledge of risks/benefits)	1. How would you assess the risk of post-operative nausea and vomiting for this patient? 2. If this patient tells you of a drug allergy, how would you proceed? 3. This patient reports having a runny nose that started 3 days ago a. What further questions would you like to ask? b. How would you differentiate between upper respiratory tract infection and allergic rhinitis? c. How would you counsel this patient with regards to anaesthetic risk? 4. Can you describe one risk stratification tool for patients undergoing surgery? 5. How would you ensure the transfer of information to the relevant healthcare team with regards to this patient's care e.g., dental work, need for further evaluation/ investigations? 6. What are the different types of anaesthetic modalities and how would you counsel a patient with regards to the benefits and risks for each modality?
EBD	Assessment of a patient with pre-existing respiratory disease	1. Relevant clinical findings or investigations to assess risk of pulmonary complications; discussion of investigations aimed at assessing lung function; optimisation strategies of various respiratory diseases such as asthma, chronic obstructive pulmonary disease (COPD), pulmonary hypertension 2. Risk assessment and optimisation of a patient with recent or ongoing lower respiratory tract infection 3. Role of prehabilitation 4. Optimisation strategies for smokers	1. Let us say the patient with COPD whom you are reviewing in clinic is planned for an elective procedure and he reports a recent reduction in effort tolerance. How would you manage this situation? 2. What are the anaesthetic considerations for a patient with GOLD stage 2 COPD undergoing elective surgery? 3. If this patient had asthma, how would you determine if the asthma control is optimised? 4. What do you know about prehabilitation and the evidence for its use? Do you

			know how to make arrangements for your patient to undergo prehabilitation? 5. If this patient had a recent lower respiratory tract infection, how would you optimise the patient for urgent surgery? 6. If your patient were a smoker who is keen to reduce the risk of perioperative respiratory complications, how would you counsel with regards to smoking cessation? 7. What do you know about vaping and its implications for anaesthesia?
EBD	Assessment of a patient with pre-existing cardiovascular disease	1. Assessment and optimisation of various cardiac conditions such as hypertension, ischemic heart disease 2. Knowledge of guidelines for further cardiac investigations and referrals e.g., new ECG changes, rhythm disorders 3. Management of a patient with a recent myocardial infarction 4. Perioperative management of patients on anticoagulants/antiplatelet agents 5. Effect and pre-procedural treatment of anaemia	1. How would you manage if this patient presented to you in the pre-anaesthetic clinic with a blood pressure of 180/110mmHg? 2. What risk assessment tools do you know that help stratify a patient's cardiac risk and algorithms that guide the need for further workup in a patient with pre-existing heart disease? 3. If this patient were on aspirin/ apixaban or warfarin, what would your perioperative management of these agents be? 4. If this patient's Hb were 9.5g/dl and you reviewed this in the pre-anaesthetic clinic, what would you do next?
EBD	Assessment of a patient with pre-existing endocrine disease	1. Assessment of complications and optimisation of the endocrine disease in the patient being discussed 2. Perioperative steroid coverage for patients with known or possible adrenal insufficiency 3. Perioperative management of patients with poorly controlled diabetes/ perioperative management of patients on various hypoglycaemic agents 4. Perioperative management of patients with thyroid disease	1. How would you assess and manage a patient with insulin dependent diabetes undergoing elective surgery? 2. If this patient were on metformin or SGLT inhibitors, how would you counsel the patient with regards to taking these medications in the perioperative period? 3. If this patient were on prednisolone 10mg om for 3 weeks what do the guidelines say about perioperative steroid coverage? 4. What are the potential complications of patients with un-optimised hyperthyroidism/ hypothyroidism undergoing surgery?

EBD	Assessment of geriatric patient	1. Assessment of geriatric patients that includes comprehensive evaluation of comorbidities, with emphasis on neurocognitive function, mental health wellbeing, sensory impairment, substance abuse, nutritional status and frailty, presence of advance care directives and end-of-life goals/plans 2. Neurocognitive assessment such as mini-cog for dementia; assessment of decision-making capacity 3. Evaluating baseline and current functional status by screening for activities of daily living and limitations in mobility such as using the Time Up and Go test (TUG) 4. Knowledge of validated scoring systems for frailty and nutritional status; knowledge of strategies for optimisation 5. Knowledge and management of chronic medications/ polypharmacy in perioperative period 6. Knowledge of factors contributing to postoperative delirium and strategies for its mitigation	1. Define a geriatric patient 2. What issues are geriatric patients more likely to face in the perioperative period and bearing this in mind, how would you counsel for anaesthesia or take informed consent? 3. What is frailty and what tools do you know of to assess this? 4. Are there interventions in the pre-operative period that may reduce the risk of postoperative delirium?
EBD	Assessment of morbidly obese patient (same patient can be used for EPA 2 DOPS mask ventilation/LMA or mask ventilation/ ETT insertion for obese patient)	1. Screening morbidly obese patients for anaesthetic risk factors such as difficult airway, technical procedural difficulties, and optimising patient for conditions associated with obesity 2. Evaluation and optimisation of associated conditions such as metabolic syndrome, cardiovascular disease, respiratory disease, endocrine disease, liver disease, gastroesophageal reflux disease and venous thromboembolism 3. Screening for obstruction sleep apnoea (e.g., STOP-BANG score) and obesity hypoventilation syndrome	1. How would you screen for obstructive sleep apnoea (OSA) preoperatively? 2. What is your preoperative management of a patient with known OSA coming for elective surgery? How would you ensure the patient is optimised for surgery? 3. What are the anaesthetic concerns for a morbidly obese patient? 4. How would you perform preoperative anaesthetic counselling for a morbidly obese patient?

Anaesthesiology EPA 2

Title	Providing anaesthetic care for adult patients requiring general anaesthesia (GA)
Specifications and limitations	**Specifications**: This EPA includes providing anaesthetic care for patients of all physiological and pathophysiological states undergoing elective and emergency procedures within the operating rooms that are otherwise not specified in the list of limitations below. 1. Ensuring all medico-legal requirements related to the procedure and anaesthesia, such as informed consent and documentation, are fulfilled. 2. Formulating the general anaesthetic plan and prioritising tasks to carry out plan. 3. Communicating information effectively to all members of the healthcare team such as during a team brief/ huddle. 4. Ensuring all preparations required to execute the plan are in place including the operating room, equipment, staff, and the employment of personal protective measures. 5. Complying with patient safety practices such as peri-procedural checklist/ timeout. 6. Managing increased risk of aspiration from delayed gastric emptying with measures such as rapid sequence induction. 7. Using appropriate physiological monitors to guide patient care. 8. Inducing and maintaining an appropriate state of balanced general anaesthesia. 9. Managing the airway of patients at all levels of consciousness. 10. Minimising harm to the patient, from iatrogenic causes or due to the anaesthetised state, procedure or underlying medical condition, including (not exhaustive) eye injuries, pressure injuries, burns, neuropathy due to positioning, hypotension due to medication, hypercarbia due to pneumoperitoneum, hypoxia due to one-lung ventilation, etc. 11. Recognising and managing peri-procedural complications/crisis in a timely manner including calling for help appropriately. 12. Reversing the state of general anaesthesia and managing the emergence of the patient. 13. Handing over care of the patient appropriately to fellow anaesthetists, nursing staff in the post-anaesthesia care unit or staff in the intensive care unit. 14. Reporting adverse events when indicated.
	Limitations: This EPA excludes the following: 1. Patients pregnant with a viable foetus. 2. Patients aged less than 16 years' old. 3. Patient undergoing procedures involving the heart and/or access to heart chambers and/or cardiopulmonary bypass or equivalent. 4. Patients undergoing procedures NOT within the operating rooms.

Sub competencies relevant for this EPA

	PC1	✓		PC7	✓		SBP1	✓		P1	✓	
	PC2	✓		PC8	✓		SBP2	✓		P2	✓	
	PC3	✓		PC9			SBP3	✓		P3		
	PC4	✓		PC10			PBL1	✓		ICS1	✓	
	PC5	✓		MK1	✓		PBL2	✓		ICS2	✓	
	PC6			MK2	✓					ICS3	✓	

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment

K:

1. Applied physiology relevant to maintaining the stable physiological state to minimise harm to the anaesthetised patient
2. Applied pharmacology relevant to maintaining the state of balanced general anaesthesia with pain relief and the stable physiological state to minimise harm to the anaesthetised patient
3. Implication of breastfeeding on administration of common anaesthetic and analgesic drugs
4. Principles, application and troubleshooting of equipment used for anaesthetic care
5. Applied anatomy relevant to perform procedures required for safe conduct of anaesthesia
6. Management of acute and chronic medical conditions pertinent to the provision of general anaesthesia in the perioperative period
7. Anaesthetic considerations regarding emergency procedures
8. Surgical/procedurals plans, guidelines and protocols associated with the peri-procedural care of the patient, such as enhanced recovery after surgery (ERAS), deep venous thrombosis prophylaxis etc.
9. Diagnosis and management of adverse conditions, including (not exhaustive) patient specific conditions such high airway pressure, malignant hyperpyrexia, thyroid storm, myocardial ischemia, arrhythmias, hypotension, hypoxaemia, changes in neurological status; equipment or facility related condition such as ventilator failure, power failure, fire, that may arise during the periprocedural period
10. Local system-based practice relevant for peri-procedural care of the patient, including informed consent, adaptations of the WHO surgical safety checklist, required documentations, handover procedures
11. Institution specific adverse events/critical incident reporting systems

Additional:

For geriatric patients

1. Physiological and pathophysiological changes associated with normal aging and frailty relevant to provision of anaesthesia
2. Differences in pharmacokinetic and pharmacodynamic effects of medications commonly used in the perioperative period in geriatric patients compared to young healthy adults
3. Drug interactions between polypharmacy for chronic diseases and anaesthetic agents

		4. Adverse perioperative events commonly associated with elderly patients such as postoperative cognitive dysfunction, postoperative delirium, skin injuries, cardiovascular or neurological ischemia 5. Neurocognitive assessment such as mini-cog for dementia 6. Medico-legal implications of neurocognitive impairment and consent for anaesthesia and surgery 7. Role of rehabilitation 8. Strategies to minimise postoperative cognitive dysfunction/ delirium **For bariatric patients** 1. Definition of morbid obesity 2. Conditions associated with morbid obesity including metabolic syndrome, cardiovascular disease, respiratory disease, endocrine disease, liver disease, gastroesophageal reflux disease and venous thromboembolism 3. Obstruction sleep apnoea and obesity hypoventilation syndrome: implications for anaesthesia and perioperative management 4. Anatomical, physiological, and pathophysiological differences associated with morbid obesity relevant to provision of anaesthesia. 5. Differences in pharmacokinetic and pharmacodynamic effects of medications commonly used in the perioperative period in obese patients compared to healthy adults. 6. Specific equipment designed for morbidly obese patients including operating table, Troop elevation pillow etc. 7. Adverse perioperative events commonly associated with morbidly obese patients 8. Anaesthetic considerations for bariatric surgery 9. Physiological implications and anaesthetic considerations for laparoscopic surgery **For intraoperative neurophysiological monitoring (IOMN) which is used in surgeries like neurosurgery, ENT surgery, spinal surgery to avoid injuries to nerves** 1. Various modes of IOMN and rationale for their use intraoperatively 2. Applied physiology and pharmacology relevant to maintaining the state of balanced general anaesthesia while facilitating IONM. 3. Principles of troubleshooting when there is a decrease or loss of signal intraoperatively **For thoracic surgery requiring lung isolation** 1. Lung function tests and diagnostic imaging of respiratory system relevant to anaesthetic management 2. Airway anatomy 3. Respiratory disease and their management implications for anaesthesia 4. Physiology of one lung ventilation 5. Indication for one lung ventilation 6. Methods of establishing one lung ventilation and managing complications such as hypoxia and impaired gaseous exchange associated with it 7. Anaesthetic considerations for pneumonectomy/ lobectomy 8. Anaesthetic considerations for thoracoscopic surgery

		For neurosurgery 1. Basic neurophysiology such as cerebral blood flow (CBF) and factors that affect it, including cerebral blood flow autoregulation, cerebral blood flow response to CO2 and cerebral metabolism 2. Pathophysiological effects of brain injury and techniques of preventing or limiting brain injury including managing raised intracranial pressure (ICP), maintaining cerebral oxygenation and perfusion, decreasing cerebral oxygen requirements such as seizure prevention or sedation 3. Effects of anaesthetic induction and maintenance agents on ICP and CBF 4. Anaesthetic considerations for neurosurgical procedures
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Management of airway • Intravascular access (venous/ arterial) • Insertion of devices within the nasal, oropharyngeal, oesophageal, and gastric cavity for the purpose of monitoring, extraction of gases or bodily fluids and/or infusion of gases or fluids 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Clinical rotation to operating theatre 2. Service updates on equipment used for the provision of anaesthetic care 3. Case-log of patients presenting for various elective and emergency procedures requiring general anaesthesia 4. Teaching sessions for directed skills acquisition in areas of airway management and central venous access 5. Teaching sessions on surgery specific anaesthetic considerations

<table>
<tr><td></td><td colspan="3">6. Courses or teaching sessions on patient safety and risk management systems, such as the Institute for Healthcare Improvement (IHI) courses or equivalent</td></tr>
<tr><td>Potential risks in case of failure</td><td colspan="3">1. Increase in patient morbidity/dissatisfaction/ mortality
2. Increase in medico-legal liability
3. Increase in healthcare costs
4. Increase in overall duration of surgery leading to delay in care of other patients</td></tr>
<tr><td rowspan="3">Sources of information to support summative entrustment decisions</td><td>Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed
(To consider evaluators and context)</td></tr>
<tr><td>Mini-Clinical Examination (Mini-CEX)</td><td>Minimum 2 by end R1
(For level 3 supervision)</td><td>Physician Faculty
To be completed by end R1
Minimum 1 each of the following
1. Patient requiring GA with endotracheal tube (inclusive of entrustment discussion on management of ventilation or oxygenation related anaesthetic crisis during general anaesthesia)
2. Patient requiring GA with LMA (inclusive of entrustment discussion on management of crisis related to anaesthetic agent use e.g., anaphylaxis, malignant hyperthermia)</td></tr>
<tr><td>Direct Observation of Procedural Skills (DOPS)</td><td>Minimum 3 of specified DOPS reaching minimal score of 5 for overall assessment.

(To note that mask holding and LMA will be assessed with the same form; same with mask holding and endotracheal intubation)</td><td>Physician Faculty
To be completed by first 2 months of R1
<u>Airway Management Skills</u>:
Minimum 1 each of the following:
1. Mask holding and bag ventilation
2. Successful LMA insertion
3. Successful endotracheal intubation</td></tr>
<tr><td></td><td>Minimum 4 of specified DOPS reaching minimal score of 5 for overall assessment</td><td>Physician Faculty
To be completed by end R2
Airway management skills in a morbidly obese patient:
Minimum 1 each of the following
1. Mask holding and bag ventilation
2. Successful endotracheal intubation</td></tr>
</table>

			To be completed by end R2 1. Placement of double lumen tube and confirmation using both auscultation and bronchoscopy
	Entrustment-based Discussion (EBD)	Minimum 1 by end R1 (For level 3 supervision)	Physician Faculty co-managing case *To be completed by end R1* 1. Management of cardiovascular related anaesthetic crisis during general anaesthesia
		Minimum 6 by end R3 (For level 4 supervision)	*To be completed by end R2* Minimum 1 each of the following: 1. Management of ASA 3 or 4 patient presenting for elective or emergency procedures requiring general anaesthesia, e.g., high risk cardiac patient presenting for non-cardiac surgery 2. Management of geriatric patient 3. Management of morbidly obese patient 4. Management of patient requiring intraoperative neurophysiological monitoring
			To be completed by end R3 Minimum 1 each of the following: 1. Management of patient undergoing thoracic surgery requiring one lung ventilation 2. Management of patient undergoing neurosurgical procedure
	Competency Evaluation	Minimum 2 per year	Physician Faculty End of anaesthesia rotation (6 monthly) evaluation on competency
	Multisource 360° Feedback (MSF)	Minimum 1 per year	Minimum 2 peers, 2 nurses, 2 senior doctors
	Case Logs	All patients undergoing GA • Geriatric patients aged 65 years and above (minimum 30)	All cases managed by resident should ideally be logged and reviewed by programme on a regular basis.

		• Morbidly obese patients (minimum 10) • Surgery requiring IONM (minimum 5) • Thoracic surgery requiring OLV (minimum 10) • Neurosurgical cases (minimum 15 of which minimum of 10 are emergency cases)	In particular, it must be ensured that minimum numbers as stipulated are met to allow the appropriate entrustment level to be assigned
Which supervision level when?	Level 3 supervision by end R2 Level 4 supervision by end R3		
Expiry	If not practised, after how many months or years will more close supervision be required again?	1 year	

Entrustment Questions for EPA 2

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Patient requiring GA with endotracheal tube	1. Ensures all medico-legal requirements/ patient safety checks have been conducted before induction of GA 2. Able to induce and maintain and reverse general anaesthesia in patient 3. Able to intubate and extubate the patient safely 4. Knows how to manage ventilation and oxygen related crisis during anaesthesia 5. Performs a systematic handover of care to PACU	1. After intubation, how do you decide what ventilator settings to set/ justify your ventilation settings for this patient? 2. When would you decide to use rapid sequence induction, and can you describe how you would conduct it? 3. What is the role of cricoid pressure and demonstrate how you would show an anaesthetic assistant how to apply it. How much force is supposed to be applied? 4. If during surgery you notice that the airway pressure keeps increasing or the high-pressure alarm goes off, can you tell me what you would do? 5. What safety features does the anaesthetic machine have to prevent delivery of a hypoxic mixture to the patient? 6. How would you interpret if the capnography shows XXX and what would you do? 7. If you notice the end tidal CO2 rising rapidly, what would you do? What would your differentials be? 8. How would you ensure that the patient's oxygen saturation remains above 95% after extubation and in PACU?
Mini-CEX	Patient requiring GA with LMA (inclusive of entrustment discussion on management of crisis related to anaesthetic agent use e.g., anaphylaxis, malignant hyperthermia)	1. Ensures all medico-legal requirements/ patient safety checks have been conducted before induction of GA 2. Able to induce and maintain and reverse general anaesthesia in patient 3. Able to justify the use of LMA and choose an appropriate type of LMA for procedure	1. Why did you choose to use LMA in this patient and why did you choose this particular type of LMA? 2. What if after giving the antibiotic/ after the surgeon started cleaning with chlorhexidine solution, you noticed that the patient's heart rate increased, and blood pressure dropped? What would you do next? 3. How do you prevent hypothermia? Which patients would be at risk for hypothermia? 4. If during the surgery, the patient suddenly starts to breathe rapidly and a 'crowing' sound is heard, what would you do?

			5. If you notice the patient's tooth is loose after removal of LMA, how would you proceed? 2. How would you recognise malignant hyperthermia and how would you manage it?
DOPS	Mask holding and LMA insertion R1: normal patient R2: morbidly obese patient	1. Counsel patient appropriately and takes informed consent 2. Able to determine adequacy of airway patency 3. Knows how to optimise mask holding and ventilation 4. Knows appropriate indication/ contraindication for LMA in patient 5. Knows the different types of LMA available 6. Able to justify choice of LMA in this particular patient 7. Communicates with healthcare staff appropriately 8. Knows when to trouble shoot problems with insertion including change in technique, apparatus, or plan 9. Ensures appropriate depth of anaesthesia before attempting insertion 10. Knows how to assess correct placement of LMA 11. Checks for cuff pressure and knows ideal cuff pressure 12. Able to name at least 2 potential complications each for mask holding and use of airway adjuncts such as oral airway or nasopharyngeal airway 13. Able to name at least 2 complications or risks associated with LMA 14. Knows safe management of laryngospasm and regurgitation 15. For the morbidly obese patient, demonstrates realisation that ventilation and oxygenation may be challenging as seen by choice	1. Describe how you will instruct a medical student on how to hold a mask and ventilate adequately. How will you know if your medical student is ventilating adequately or excessively? 2. What are some of the complications from mask-holding or using the various airway adjuncts? 3. How will you choose the type and size of LMA for your case? 4. How would you teach a medical student to insert the LMA and how does what you describe help with the insertion? 5. How do you know that the LMA is properly inserted? 6. What to do if there is a leak detected in the circuit with the LMA in place? 7. What is the acceptable cuff pressure for the LMA? 8. What are some of the complications or risks of using the LMA? 9. You have a young patient who is listed for emergency evacuation of uterus following incomplete spontaneous abortion. You induced general anaesthesia in your patient and successfully inserted the LMA. When the patient is moved down the OT and placed into a lithotomy position, you noticed large amounts of brownish fluid welling up from the drain tube (gastric channel) of the LMA and in her mouth. What will you do now? What are the possible triggers for this? 10. If following skin incision, there is an inspiratory stridor from the patient and then no waveform on the capnogram. What has happened? What will you do now? How can you prevent this from happening?

		of LMA and technique used to ensure proper placement and assisted ventilation	
DOPS	Mask holding and ETT insertion in normal airway R1: normal patient R2: morbidly obese patient (same patient can be used for EPA 1)	1. Knows indications for ETT intubation in the patient 2. Takes appropriate informed consent 3. Does proper airway assessment and recognises a potentially difficult airway 4. Ensures required equipment is prepared before induction of anaesthesia 5. Knows the difference between a curve and straight blade and how each is supposed to be used 6. Ensures adequate hypnosis and muscle relaxation before intubation attempt 7. Ensures patient is oxygenated well before intubation attempt 8. Communicates appropriately with healthcare team to facilitate good intubating condition 9. Monitors patient and demonstrates awareness of patient's response to intubation attempt 10. Demonstrates correct intubation technique 11. Knows when to abandon attempt and escalate when necessary 12. Knows how to determine correct tracheal tube placement (clinical and other methods like carbon dioxide detection) 13. Measures cuff pressure and knows ideal cuff pressure 14. In a morbidly obese patient, demonstrates proper positioning of the patient (e.g., using a troop pillow), ensures adequate pre-oxygenation and uses 2 men ventilation technique, considers apnoeic oxygenation	1. What are some types of patients who might be difficult to mask ventilate and why? 2. If morbidly obese Santa Claus (with his enormous facial hair) shows up one night when you are on call for a GA, would you expect any problems with oxygenation and how will you get around the problem (aside from getting OT technical assistant to shave him)? 3. Describe how you will instruct a medical student on how to intubate a patient with an appropriate size endotracheal tube and confirm correct position of endotracheal tube. How would you know if the endotracheal tube is at the level of the carina? 4. Can you describe at least 5 different types of endotracheal tubes that are available in your hospital? 5. Why did you choose this particular type of endotracheal tube? What are some of the advantages and disadvantages of this endotracheal tube (assessor to decide on at least one variety of the named endotracheal tubes above)? 6. What are some of the complications of endotracheal intubation? How can you minimise these complications? 7. You have a young ASA-1 patient who came for emergency laparoscopic appendectomy. You performed rapid sequence induction and successfully intubated the patient. When you tried to check the position of the endotracheal tube by auscultation, you find that you have difficulty ventilating even with high pressures. You are unable to hear any breath sounds. What are the possible causes? What will you do now?

DOPS	Placement of double lumen tube (DLT) and confirmation using both auscultation and bronchoscopy	1. Takes appropriate informed consent with counselling for generic risks associated with DLT 2. Performs adequate airway assessment 3. Prepares and checks appropriate choice of equipment for airway including size of DLT and FOB, checking of cuffs of DLT 4. Articulates reason for choice of DLT in patient for OLV 5. Demonstrates knowledge of anatomy of tracheobronchial tree 6. Able to troubleshoot equipment issues to optimise view of tracheobronchial tree 7. Appropriate analgesia/ sedation 8. Ensures good intubating conditions before attempting intubation 9. Ensures usual ASA monitors in place during procedure 10. Confirms appropriate position of tube with FOB 11. Checks correct position via auscultation 12. Demonstrates correct steps resulting in appropriate depth of insertion with minimal complications during insertion 13. Knows how to manage the FOB and diagnose/ troubleshoot problems that result in inadequate view of the tracheobronchial tree 14. Knows how to adjust if the DLT has been placed in the incorrect bronchus 15. Measures cuff pressure of tracheal tube 16. Effective communication with the healthcare team 17. Seeks help appropriately if unable to rectify problems encountered 18. Knows when to stop if the patient is not tolerating the intubation attempts	1. Can you describe the difference between a left and right sided DLT? 2. How would you insert a right DLT and what do you need to note when using a right sided DLT? 3. Intraoperatively, the patient desaturates to SpO2 80%. Describe your immediate management of this patient. 4. If the patient show signs of difficult airway e.g., receding chin with limited mouth opening and neck extension, what are your concerns prior to insertion of DLT? Are there any airway adjuncts/equipment that you would like to standby prior to induction?

EBD	Management of cardiovascular related anaesthetic crisis during general anaesthesia	1. Differential diagnosis and management of hypotension/ hypertension/arrhythmias intraoperatively	1. If this patient's blood pressure falls to 70/30mmHg the next time the NIBP cycles, what would you do? 2. Would your approach be different if this patient's heart rate increases to 120/min versus 190/min? How would you manage the patient
EBD	Management of ASA 3 or 4 patient presenting for elective or emergency procedures requiring general anaesthesia, e.g., high risk cardiac patient presenting for non-cardiac surgery	1. Anaesthetic management of a patient with high cardiac risk (e.g., triple vessel disease that has not been revascularized, ischemic cardiomyopathy with poor ejection fraction, severe aortic stenosis, right ventricular dysfunction) presenting for non-cardiac surgery 2. Anaesthetic management of a patient with a multi-systemic disease presenting for surgery e.g., rheumatoid arthritis or other autoimmune disease/ metastatic cancer/endocrine disease	1. How will you conduct general anaesthesia for a patient with high cardiac risk for non-cardiac surgery? 2. Can you explain the anaesthetic considerations for a patient with high cardiac risks and the rationale for your anaesthetic management? How do you optimise the myocardial oxygen supply and demand in a patient with high risk of myocardial ischemia? 3. Describe your perioperative fluid management plan for a patient with congestive cardiac failure. What are the anaesthetic considerations for a patient with severe valvular heart disease e.g., severe aortic stenosis? 4. After induction of anaesthesia, you noted that the patient has developed ST depression in lead II on your anaesthesia monitor. Describe how would you manage this patient. 5. Intraoperatively, you noted that this patient developed fast AF, HR 130s, BP 80/40. Describe your immediate management. 6. How would you manage a patient on disease modifying anti-rheumatic drugs? 7. What precautions would you take when positioning the patient with multi-joint arthritis?
EBD	Management of geriatric patient	1. Modification of anaesthetic technique taking into account underlying physiological or pathophysiological changes 2. Optimising provision of GA using monitors like depth of anaesthesia monitoring, regional cerebral oxygen saturation and neuromuscular monitoring	1. What is the definition of a geriatric patient? 2. How would your anaesthetic for this geriatric patient differ from a patient who is 40 years old without significant medical history? 3. How can we reduce the risk of post-operative cognitive

		3. Prevention of postoperative cognitive dysfunction, postoperative delirium, skin injuries, cardiovascular or neurological ischemia	delirium/dysfunction/stroke in this patient through our intraoperative management? 4. If this patient had moderate dementia, how would your anaesthetic technique differ? 5. If the PACU nurse calls you to say this patient is agitated and trying to climb out of the trolley, what would you do? 6. What instruments for assessment of delirium do you know of?
EBD	Management of morbidly obese patient	1. Modification of techniques for patients with suspected or confirmed obstructed sleep apnoea (OSA) 2. Positioning challenges and prevention of position related injuries (staff and patient) 3. Management of complications like numbness resulting from poor positioning 4. Anaesthetic considerations for bariatric surgery 5. Anaesthetic considerations for laparoscopic surgery 6. Postoperative disposition of patient	1. If this patient has severe OSA, how would your anaesthetic technique be different from if he/she does not have OSA? 2. What other intraoperative considerations are required for this patient? What precautions should be taken with regards to positioning? How would you manage anaesthetic drug doses? 3. How is obesity defined? 4. What are the different types of bariatric procedures that can be performed? What are the criteria for undergoing bariatric surgery? Do you know of any screening tool that would help identify the obese patient at risk of postoperative complications that would benefit from ICU/HD care? 5. Intraoperatively, what problems do you anticipate with pneumoperitoneum being created during laparoscopy? How would you manage ventilation in the morbidly obese patient in a head down position?
EBD	Management of patient requiring intraoperative neurophysiologic al monitoring	1. Describing the various modes and indications of Intraoperative Neurophysiological Monitoring (IONM) 2. Adapting anaesthetic management to facilitate various modes of Intraoperative Neurophysiological Monitoring (IONM) 3. Troubleshooting loss of signal from IONM	1. How would you conduct anaesthesia for a surgery that requires IOMN e.g., SSEP (somatosensory evoked potential) MEP (motor evoked potential), BAEP (Brainstem auditory evoked potential) EMG (electromyography)? 2. What are the various modes of IONM and their indications for use in surgery? E.g., How are SSEP and MEP performed and what do they monitor intraoperatively? 3. What are the anaesthesia factors that will affect the IONM readings intraoperatively? Can we use analgesic agents e.g., ketamine

			or dexmedetomidine in a surgery with IONM? 4. What would you do if there is loss of IONM signals intraoperatively? 5. Would your management be different if the signal loss is unilateral or bilateral?
EBD	Management of patient undergoing thoracic surgery requiring one lung ventilation	1. Establishing the need for one lung ventilation 2. Choice of lung isolation technique for surgery 3. Management of complications during one lung ventilation such as hypoxia, bronchospasm 4. Anaesthetic considerations for pneumonectomy/ lobectomy/thoracoscopic surgery including choice of anaesthetic agents, fluid management, pain management 5. Post thoracic surgical complications to watch out for 6. Extubation strategy	1. Describe how will you conduct anaesthesia for a patient coming for a. A video-assisted thoracotomy (VAT) procedure? b. Open lung lobectomy? c. Open pneumonectomy? How will you perform the lung isolation in each of these procedures? 2. What are the physiological changes during one lung ventilation? 3. Describe your plans and rationale for fluid management during anaesthesia for a thoracic surgery 4. What are the analgesic options for this thoracic surgery? Justify your choice. 5. What are the early post-operative complications that you will look out for after a pneumonectomy? 6. After positioning and lung isolation, if the patient desaturates to SpO2 85%, what are your differentials and what would you do? 7. How would you perform lung isolation in a patient with a known difficult airway presenting for a VATS procedure?
EBD	Management of patient undergoing neurosurgical procedure	1. Intraoperative management of raised intracranial pressure and ensuring cerebral perfusion, oxygenation and metabolic requirements are met 2. Additional anaesthetic risks associated with emergency procedures such as aspiration risks and limited time for optimisation 3. Appropriate disposition for the patient post-operatively 4. Handing over of care of the patient appropriately including to fellow anaesthetist, nursing staff	1. How would you provide anaesthesia for an emergency neurosurgery for a patient with raised intracranial pressure? 2. Can you explain the goals of anaesthesia for this neurosurgery? How does your anaesthetic technique help to achieve this? 3. The surgeon inserts an ICP monitor intraoperatively, and it reads 40mmHg. Describe your subsequent anaesthetic management to treat this raised ICP. 4. If the patient requires an urgent procedure at the neuroangio suite (remote location), how would you organise the transport and safe conduct of anaesthesia for a patient

		in post-anaesthesia care unit or staff in the intensive care unit 5. Management of patient with raised intracranial pressure presenting for neurosurgical procedures with a focus on specific issues such as management of raised intra-cranial pressure, ensuring sufficient cerebral perfusion and minimising secondary brain injury 6. Additional considerations if patient requires transport to radiology suite	with depressed GCS and raised ICP at a remote location? 5. If the surgeon requires to perform decompressive craniectomy in a prone position, what are your additional anaesthetic considerations?

Anaesthesiology EPA 3

Title	**Providing anaesthetic care for patients under regional anaesthesia (RA)**
Specifications and limitations	**Specifications**: 1. Assessing patient suitability for RA including evaluating baseline neurological deficits 2. Formulating a regional anaesthetic plan with consideration for the type of surgical procedure, and the risks/benefits of various options including local anaesthesia, peripheral nerve blocks (PNB), central neuraxial blocks, sedation and monitored anaesthetic care (MAC) 3. Utilising standard monitoring equipment, procedure-specific equipment, and drugs 4. Performing regional anaesthesia related patient safety checklist including ensuring correct block for the surgery to be performed, informed consent, optimal patient positioning and all other preparations in place 5. Performing RA with attention to sterility, patient comfort and proceduralist's ergonomic considerations 6. Providing monitored anaesthesia care for patients who have been given RA block by anaesthetists/ surgeon 7. Managing side effects and complications associated with RA technique including failed RA/ local anaesthetic systemic toxicity (LAST)/ neuropraxia 8. Formulating and implementing post RA care plan 9. Communicating care plans with patients and inter-professional teams 10. Documenting procedure including complications if any
	Limitations: This EPA excludes the following: 1. Patient aged less than 16 years old
Sub competencies relevant for this EPA	

	PC1	✓		PC7	✓		SBP1	✓		P1		
	PC2	✓		PC8	✓		SBP2	✓		P2	✓	
	PC3	✓		PC9			SBP3	✓		P3		
	PC4	✓		PC10	✓		PBL1	✓		ICS1	✓	
	PC5			MK1	✓		PBL2	✓		ICS2	✓	
	PC6			MK2	✓					ICS3	✓	

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	**1. Basic science/ General** • Anatomy of nervous system and eye • Neuromuscular physiology, haemostatic physiology • Physiological changes and monitors used during sedation • Pharmacology related to local anaesthetics and adjuncts, sedation medications, pharmacokinetics of drugs in epidural/subarachnoid space, • Physics and equipment for safe and accurate delivery of local anaesthetics (including ultrasound machine, needles, peripheral nerve stimulator) **2. Guidelines for:** • Management of local anaesthetic systemic toxicity (LAST) • Neurological complications associated with RA technique • RA technique in patients receiving anticoagulant/antiplatelet/antithrombotic agents **3. RA technique specific**: • General considerations and equipment for specific RA technique • Applied anatomy: surface anatomy, sonoanatomy for relevant blocks • Side effects and complications of RA technique • Types of blocks: I. Central neuraxial: subarachnoid block, epidural, caudal II. Cervical plexus III. Upper limb: brachial plexus (interscalene, supraclavicular, infraclavicular/ costoclavicular, axillary) IV. Lower limb: lumbar plexus, fascia iliaca, femoral nerve, adductor canal, saphenous nerve, obturator nerve, sacral plexus, sciatic nerve, ankle block V. Truncal: transversus abdominis plane, rectus sheath, iliohypogastric/ ilioinguinal, paravertebral, erector spinae plane, intercostal VI. Others: dorsal penile nerve, Bier's block VII. Ophthalmic blocks: subtenon, peribulbar, retrobulbar **4. Surgery-related considerations:** • Common surgical procedures performed under RA, including ophthalmic surgery, orthopaedic procedures with or without use of tourniquet

	S:	1. Clinical assessment skills: history taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills a. Use of ultrasound machine (image acquisition, optimisation, interpretation), needles, catheters, peripheral nerve stimulator, continuous infusion devices b. Ability to correlate surface anatomy with sonoanatomy c. Perform specific RA blocks: d. **Central neuraxial blocks**: subarachnoid block, epidural block e. **Brachial plexus blocks**: interscalene/ supraclavicular/infraclavicular/axillary f. **Lower limb nerve blocks**: i. Anterior compartment e.g., Fascia Iliaca/femoral nerve block ii. Posterior compartment e.g., sciatic block g. **Truncal blocks**: erector spinae plane/rectus sheath/iliohypogastric, ilioinguinal nerve/paravertebral/ transversus abdominus plane block 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides patient-centric compassionate care, ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Clinical rotations: • Multispecialty anaesthesia rotations/core anaesthesia training that cover orthopaedic, general, urological, thoracic, vascular, gynaecology and ophthalmic surgeries. • Obstetric anaesthesia rotation (at least 4 months in Junior Residency (R1 to R3) and 2 months in Senior Residency (R4 to R5) • RA rotation during Junior Residency (R1 to R3) with a focus on peripheral nerve blocks (at least 1 month) 2. Case Logs 3. Teaching sessions

<table>
<tr><td>Potential risks in case of failure</td><td colspan="3">1. Increase in medico-legal liability
2. Increase in healthcare cost
3. Increase in patient dissatisfaction/ morbidity/ mortality
4. Conversion to general anaesthesia
5. Increase in overall duration of surgery leading to delay in care of other patients</td></tr>
<tr><td rowspan="4">Sources of information to support summative entrustment decisions</td><td>Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed
(To consider evaluators and context)</td></tr>
<tr><td>Mini-Clinical Examination (Mini-CEX)</td><td>Minimum 1 by end R2)
(For level 3 supervision)</td><td>Physician Faculty
To be completed by end R2
1. Manage a patient for surgery under peripheral nerve block and sedation</td></tr>
<tr><td>Direct Observation of Procedural Skills (DOPS)

(The same case may be used for the relevant EBD discussion for this EPA; Please note that 2 different forms i.e., DOPS and EBD will have to be filled)</td><td>Minimum 5 (1 each) reaching minimal score of 5 for overall assessment.</td><td>Physician Faculty
To be completed by end R1
1. Subarachnoid block
To be completed by end R2
2. Upper limb block (single shot) Brachial plexus
3. Lower limb block (single shot)
Any one of the following blocks:
• Fascia iliaca,
• Femoral nerve block,
• Popliteal approach to sciatic nerve block
To be completed by end R3
4. Truncal block (single shot)
Any one of the following blocks
• Erector spinae block,
• Rectus sheath block,
• Transversus abdominis plane block,
• Ilioinguinal/ iliohypogastric nerve block
To be completed by end R5
5. Peripheral nerve block with catheter</td></tr>
<tr><td>Entrustment-based Discussion (EBD)</td><td>Minimum 1 by end R1
(For level 3 supervision)</td><td>Physician Faculty co-managing case
To be completed by end R1
(For level 3 supervision)
1. Management of a patient undergoing ophthalmic surgery under MAC</td></tr>
</table>

		Minimum 1 by end R2 (For level 3 supervision)	Physician Faculty co-managing case *To be completed by* end R2 (For level 3 supervision) 2. Management of a patient undergoing surgery under subarachnoid block
		Minimum 1 by end R3 (For level 4 supervision)	Physician Faculty co-managing case *To be completed by* end R3 (For level 4 supervision) 3. Management of a patient undergoing surgery under subarachnoid block/single shot peripheral nerve block
		Minimum 1 by end R3 (For level 3 supervision)	Physician Faculty co-managing case *To be completed by* end R3 (For level 3 supervision) 4. Management of patient undergoing surgery with a peripheral nerve block catheter
		Minimum 1 by end R5 (For level 4 supervision)	Physician Faculty co-managing case *To be completed by* end R5 (For level 4 supervision) 5. Management of patient undergoing surgery with a peripheral nerve block catheter
	Competency Evaluation	Minimum 1	End of RA rotation
	Multisource 360° Feedback (MSF)	Minimum 2	Setting where RA is performed • Minimum 1 nurse and 1 physician faculty • Minimum 1 in Junior Residency (R1 to R3) • Minimum 1 in Senior Residency (R4 to R5)
	Case Logs	Minimum cumulative numbers (40) • 10 by R2 • 20 by R3 • 30 by R4 • 40 by R5	Reviewed by programme Peripheral nerve blocks • 10 upper limb RA blocks • 10 lower limb RA blocks • 5 truncal blocks Minimum numbers must be achieved each year to qualify for: • Level 3 supervision by end R2 • Level 4 supervision by end R3
		Minimum cumulative numbers (40) • 10 by R2 • 20 by R3 • 30 by R4 • 40 by R5	• Subarachnoid blocks o Combined spinal epidural (CSE) can be logged under both subarachnoid block and epidural. o Obstetric subarachnoid blocks can be included

<table>
<tr><td rowspan="2"></td><td rowspan="2"></td><td>Minimum cumulative numbers (40)
• 10 by R2
• 20 by R3
• 30 by R4
• 40 by R5</td><td>• Epidural/ caudal
o Combined spinal epidural (CSE) can be logged under both subarachnoid block and epidural.
o Labour epidural for obstetric patients can be included</td></tr>
<tr><td>Minimum cumulative numbers (5)
• 1 by R3
• 5 by R5</td><td>Peripheral nerve block with catheter
Minimum numbers must be achieved to qualify for:
• Level 3 supervision by end R3
• Level 4 supervision by end R5</td></tr>
<tr><td>Which supervision level when?</td><td colspan="3">For patients under RA (including single shot peripheral nerve blocks and subarachnoid blocks)
• Level 3 supervision by end R2
• Level 4 supervision by end R3

For patients under RA with catheter
• Level 3 supervision by end R3
• Level 4 supervision by end R5

Resident is expected to exit the programme at entrustment level 4</td></tr>
<tr><td>Expiry</td><td colspan="2">If not practiced, after how many months or years will more close supervision be required again?</td><td>1 year</td></tr>
</table>

Entrustment Questions for EPA 3

Assessment Tools	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Manage a patient for surgery under peripheral nerve block and sedation	1. Sedation options 2. Monitoring required 3. EBD discussion points as above for DOPs	1. Explain your choice of sedation 2. what would you do if 30 min later, the patient starts to get restless?
DOPS	Spinal/ subarachnoid block	1. Contra-indications to block including management of anticoagulants/ antiplatelets/ antithrombotic agents 2. Alternative approaches to performance of block 3. Choice of local anaesthetic and dose 4. Patient safety checklist	1. What are the contraindications to subarachnoid block? 2. If this patient were on aspirin/clopidogrel/apixaban, how would you proceed with the spinal? 3. If you encounter difficulty getting into the subarachnoid space with this needle and keep hitting bone, how would you modify your technique? 4. Explain your choice of spinal needle
DOPS	Upper limb block (single shot): Brachial Plexus	1. Contra-indications to block 2. Alternative approaches to performance of block e.g., addition of peripheral nerve stimulator 3. Post block care instructions	1. How would you use a nerve stimulator to aid in performing this block? 2. How would you recognise Horner's syndrome/ diaphragmatic paralysis/ pneumothorax? 3. If patient develops incomplete block, paraesthesia/ Horner's syndrome/ diaphragmatic paralysis/ pneumothorax, what would you do?
DOPS	Lower limb block (single shot) Any one of the following blocks: • Fascia iliaca • Femoral nerve block • Popliteal approach to sciatic nerve block	1. Contra-indications to block 2. Alternative approaches to performance of block e.g., addition of peripheral nerve stimulator 3. Post block care instructions	1. If the patient develops foot drop that is noted during pain round the next day, how would you proceed in managing the patient? 2. If the patient complains of persistent numbness, what would you do and how would you counsel the patient 3. If a 40kg patient had 0.5% 20mls of bupivacaine given each for a femoral nerve block and sciatic nerve block and complains of

			tinnitus and giddiness soon afterwards, what would you do?
DOPS	Truncal block (single shot) Any one of the following blocks: • Erector spinae block • rectus sheath block • Transversus abdominis plane block • ilioinguinal/ iliohypogastric nerve block	1. Indications and contraindications for block 2. Alternative to block	1. What are the signs and symptoms of local anaesthetic systemic toxicity? 2. How would you provide satisfactory surgical conditions if a patient wanted to do ilioinguinal herniorrhaphy under local anaesthetic only?
DOPS	Any peripheral nerve block with catheter	1. Indications and contra-indications to block and use of catheter 2. Alternative approaches to performance of block including additives 3. Care of catheter and troubleshooting catheter related problems	1. If the patient develops pain from the nerve block which is not working satisfactorily intraoperatively, what would you do? 2. If the patient develops neuropraxia post catheter removal, how would you manage? 3. If there is difficulty flushing the catheter, what would you do? 4. The patient/ ward staff note that there is leaking of fluid around the injection site, how would you manage this? 5. When removing the catheter, what should you look out for? 6. What instructions does the patient need to be told with regards to be having the catheter in place for post-operative pain relief?
EBD	Management of a patient undergoing ophthalmic surgery under MAC	1. Sedation options 2. Monitoring required	1. If the patient starts to panic when the drapes go over face, what would you do? 2. If the surgeon wants to give a peribulbar block for the eye and as he does so, bradycardia develops.

			What do you need to consider and proceed to do?
EBD	Management of a patient undergoing surgery under subarachnoid block	1. Sedation options 2. Monitoring required 3. choice of local anaesthetic and dose	1. How would you recognise a high block/ total spinal? 2. Are there alternatives to spinal block for this patient? 3. Patient develops hypotension/ inadequate block/high block/total spinal/post dural puncture headache, what would you do?
EBD	Management of a patient undergoing surgery under subarachnoid block/single shot peripheral nerve block	1. Sedation options 2. Monitoring required 3. Alternative approaches to performance of block e.g., addition of peripheral nerve stimulator 4. Post block care instructions	1. What if the patient under sedation becomes restless halfway through the surgery? 2. What is your analgesic plan for the patient to avoid analgesic gap?
EBD	Management of patient undergoing surgery with a peripheral nerve block catheter	1. Indications and contra-indications to block and use of catheter 2. Alternative approaches to performance of block including additives 3. Care of catheter and troubleshooting catheter related problems	1. What are the advantages of inserting a catheter for peripheral nerve block in this patient? 2. If this patient were very scared of having the block done awake, how would you proceed? 3. What post-operative care instructions would you give and how would you ensure that the information is conveyed to the patient and ward staff? 4. If you were the senior resident on call and your junior resident calls you in because there is resistance on catheter removal, how would you proceed?

Anaesthesiology EPA 4

<table>
<tr><th>Title</th><td colspan="13">Assessing and managing patients with potentially difficult airway in the hospital setting</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="13">Specifications:
1. Identifying patients with features of any of the four categories of anticipated difficult airway – bag mask ventilation, intubation, supraglottic airway, front of neck access; and assessing risk of aspiration
2. Discussing risks with the patient and obtaining informed consent
3. Planning and executing management options for ventilation, oxygenation, intubation, extubation and post-operative care of patients with potentially difficult airway including ensuring optimal patient positioning
4. Managing an awake intubation
5. Formulating back-up airway management plans in accordance with difficult airway algorithms
6. Preparing the operating theatre, equipment, drugs, and personnel required in the management of difficult airway
7. Communicating concerns and coordinating multidisciplinary care of patients with an anticipated difficult airway
8. Seeking assistance from colleagues more experienced in the management of difficult airways
9. Facilitating surgery requiring a shared airway
10. Managing difficult airway in locations outside of main operating theatre.</td></tr>
<tr><td colspan="13">Limitations:
This EPA excludes the following
1. Unanticipated difficult airway scenarios</td></tr>
<tr><th>Sub competencies relevant for this EPA</th><td></td><td>PC1</td><td>✓</td><td></td><td>PC7</td><td>✓</td><td></td><td>SBP1</td><td></td><td></td><td>P1</td><td></td><td></td></tr>
<tr><td></td><td>PC2</td><td></td><td></td><td>PC8</td><td></td><td></td><td>SBP2</td><td>✓</td><td></td><td>P2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC3</td><td>✓</td><td></td><td>PC9</td><td></td><td></td><td>SBP3</td><td></td><td></td><td>P3</td><td></td><td></td></tr>
<tr><td></td><td>PC4</td><td></td><td></td><td>PC10</td><td></td><td></td><td>PBL1</td><td>✓</td><td></td><td>ICS1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC5</td><td>✓</td><td></td><td>MK1</td><td>✓</td><td></td><td>PBL2</td><td>✓</td><td></td><td>ICS2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC6</td><td></td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td></td><td></td><td>ICS3</td><td>✓</td><td></td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	1. Difficult Airway Algorithm guidelines (e.g., Difficult Airway Society (DAS)/ ASA) for intubation and extubation, including obstetrics and paediatrics Difficult Airway Algorithms 2. Airway physiology 3. Anatomy of airway, neck, and chest 4. Interpreting airway investigations including imaging such as X-ray, CT scan and MRI to aid airway management 5. Airway management devices e.g., nasal, and oral airway, first and second generation supraglottic airways, different types of endotracheal tubes, tracheostomy tubes, flexible bronchoscope, airway exchange catheters, bougies, optical stylets, laryngoscopes 6. Devices to facilitate oxygenation e.g., high-flow nasal cannula, jet oxygenation/ ventilation 7. Techniques to optimise oxygenation through appropriate positioning, pre-oxygenation, and apnoeic oxygenation 8. Knowledge of airway equipment available in institution-specific difficult airway trolley 9. 'Front of neck access' techniques and indications 10. Pharmacology of drugs used during airway management including neuromuscular blockers
	S:	1. Clinical assessment skills • Recognising features of a potentially difficult airway by history, examination, and review of pertinent investigations • Assessing adequacy of airway patency and ventilation • Identifying and correcting problems and complications associated with airway management 2. Procedural skills • Use of airway adjuncts including nasal and oral airways • Use of supraglottic airway devices as rescue device and conduit for flexible bronchoscopy assisted intubation • Use of different intubation techniques including use of video-laryngoscopes and flexible bronchoscopy • Use of intubation adjuncts such as bougie and stylet • Use of shared airway techniques including use of micro-laryngeal tubes and jet ventilation 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)

	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides patient-centric compassionate care, ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Multispecialty anaesthesia rotation including Ear nose throat (ENT) surgery 2. Case logs for airway surgery 3. Residency programme teaching/Difficult airway training session including Cannot Intubate Cannot Oxygenate (CICO) scenario management 4. Institution code activation for emergency airway management
Potential risks in case of failure	1. Increased patient morbidity/dissatisfaction/ mortality 2. Increase in healthcare costs 3. Increased medico-legal liability	

Sources of information to support summative entrustment decisions	**Tools**	**Number to be completed satisfactorily**	**Additional specifications if needed (To consider evaluators and context)**
	Mini-Clinical Examination (Mini-CEX)	Minimum 2 by end R3 for level 3 supervision	Physician Faculty *To be completed by end R3* 1. Anaesthetic management of a patient with anticipated difficult airway 2. Anaesthetic management of a patient going for surgery with a shared airway
	Direct Observation of Procedural Skills (DOPS)	Minimum 4 by end R3 reaching minimal score of 5 for overall assessment.	Physician Faculty *To be completed by end R3* 1. Minimum 1 DOPS for awake flexible bronchoscopy guided intubation 2. Minimum 3 DOPS for difficult airway intubation using three different types of laryngoscope blades (curved, hyper-angulated, straight) and/or different models of video-laryngoscopes
	Entrustment-based Discussion (EBD)	Minimum 1 by R5 for level 4 supervision	Physician Faculty co-managing case *To be completed by end R5* 1. Patient with a difficult airway, requiring alternative intubation approaches such as awake intubation +/- front of neck access.
	Competency Evaluation	1 during airway/ ENT rotation	Physician Faculty *To be completed by end R3*

Anaesthesiology EPA 4 *(Click here to return to Table of Content)*

	Case Logs	Minimum 10	Reviewed by programme *To be completed by end R5* 1. Shared airway surgery (e.g., dental, ENT, airway instrumentation, airway tumour)
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 4

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Anaesthetic management of a patient with anticipated difficult airway	Obtains a comprehensive assessment of a potentially difficult airway • History, examination, and review of pertinent airway investigations including nasoendoscopy pictures, X-ray, CT scan and MRI • Features for each of the four categories of anticipated difficult airway - bag mask ventilation, intubation, supraglottic airway, front of neck access Understands and remembers key components of the Difficult Airway Algorithm by the Difficult Airway Society • Techniques for intubation • Bag mask ventilation • Uses of supraglottic airway • Equipment needed and steps for front-of-neck access Knowledge of options for airway management devices according to Mini-CEX scenario, the advantages and limitations, including micro-laryngeal tube and jet ventilation Formulates an airway management plan and communicates it to the relevant multidisciplinary teams • Explains choices made in management plan • Appropriately articulates anticipated concerns with the patient's airway	1. What features in this patient suggest a difficult airway/what do you think your greatest challenge is in having to manage this airway? 2. What is your management plan for this patient? 3. What if things had not gone according to your plan? 4. What circumstances would prompt you to consider an awake option for airway securement? 5. What guidelines for difficult airway management are you aware of? 6. What considerations would you have with respect to extubating this patient e.g., if this were a grade 4 larynx that you had great difficulty intubating, what would your extubation strategy be?

		• Coordinates airway management plans in a responsible and collegial manner Able to explain or justify the choices made in the airway management plan Prepares for potential complications • What if Plan A fails? • What would resident do in the event of equipment failure while managing the airway? Appropriate counselling of patient with regards to airway management concerns and plans • Demonstration of cultural competence in communication • Attention to detail in documentation Knowledge of airway equipment available in institution-specific difficult airway trolley and in airway bag Demonstrates professionalism and reflects on performance, seeks avenues for improvement.	
Mini-CEX	Anaesthetic management of a patient going for surgery with a shared airway	Objectives of anaesthesia in a shared airway procedure • Methods to safely secure airway • Surgical requirements for airway in this particular type of procedure • Options for maintenance of anaesthesia for this specific type of shared airway procedure • Considerations for extubation and techniques for safe extubation	1. Why are the benefits of your chosen anaesthetic technique for this particular patient undergoing this particular surgery? 2. What are the potential risks to the patient with regards to your chosen airway management and anaesthetic technique? 3. Are there alternative airway management plans you would consider in this patient undergoing this particular surgery?

		• If applicable, interpreting airway investigations e.g., nasoendoscopy pictures, X-rays, CT scans and MRI. • Knowledge of options for airway management devices according to Mini-CEX scenario, their advantages and limitations, including micro-laryngeal tube and jet ventilation • Formulate an airway management plan and communicate it to relevant multidisciplinary teams • Explain choices made in management plan • Appropriately articulates concerns with the patient's airway • Coordinates airway management plans in a responsible and collegial manner • Able to explain the reasoning behind choices made in airway management plan • Able to troubleshoot intraoperative airway problems • What would the resident do if the airway device gets inadvertently dislodged during surgery? Appropriate counselling of patient with regards to airway management concerns and plans • Demonstration of cultural competence in communication	4. How is jet ventilation performed? 5. Would your airway management plans change if the patient were morbidly obese with a high risk of aspiration? 6. What are the signs you would be looking out for that might suggest loss of control of the airway (e.g., airway has been dislodged), and what would you do? 7. What are the steps to take if there is sudden high airway pressure during the surgery?

		• Attention to detail in documentation Timely and responsible handover of airway issues post-operatively to relevant teams Demonstrates professionalism and reflects on performance, seeks avenues for improvement.	
DOPS	Scope assisted tracheal intubation	Indications for scope assisted intubation Preparation and utility of equipment used for awake scope assisted intubation Appropriate counselling of patient with regards to airway management concerns and plans • Demonstration of cultural competence in communication • Attention to detail in documentation Methods of airway topicalization including maximum doses Options for sedation during awake intubation Clear communication with patient throughout intubation process Demonstrates professionalism and reflects on performance, seeks avenues for improvement For detailed steps and expectations, please refer to the guide for DOPS	1. If blood obscures the scope image, what would you do and what would you tell the patient? 2. What sort of sedation would you consider appropriate for this patient? 3. What other methods of airway topicalization do you know of? 4. Why were awake scope intubation chosen for this patient and when would this technique not be a suitable choice?

DOPS	For difficult airway intubation using three different types of laryngoscope blades (curved, hyper-angulated, straight) and/or different models of video-laryngoscopes	Advantages and disadvantages of different laryngoscope models and blade types • Quality of display • Ease of obtaining view • Ease of intubation • Availability of hyper-angulated or straight blades • Portability • Disposable or non-disposable parts • Battery operated or not • Cost Laryngoscope models and blade types available in different locations in the hospital e.g., OT, ICU, radiology suites etc. Demonstrates professionalism and reflects on performance, seeks avenues for improvement.	1. Can you tell me what alternative video-laryngoscopes you know of? 2. Name 3 models of video-laryngoscopes and compare their advantages and disadvantages with what you are using today. 3. What sort of situations should different types of laryngoscope blades i.e., curved, hyper angulated, straight blade, be used? 4. How would you perform an awake intubation with a video-laryngoscope and when would you use such a technique?
EBD	Airway management of patient with a difficult airway, requiring alternative intubation approaches such as awake intubation +/- front of neck access	Decision making process – Plans A, B, C Communicating with patient – risks, informed consent Coordinating with teams – preparing OT and personnel, communicating concerns with multidisciplinary teams Extubation strategy of an 'at-risk' airway Resource and manpower management	50-year-old gentleman, 60kg, is a known case of nasopharyngeal carcinoma with previous radiation to the head and neck. He has well-controlled hypertension, cataracts, and some hearing loss, but no other chronic medical issues. He is now admitted for acute appendicitis requiring surgery. Airway assessment reveals a mouth opening of 2.5 finger-breaths and some limitation of neck movements. Previous anaesthetic records on EMR documented his larynx as Grade 1. Post rapid sequence induction, direct laryngoscopy reveals a Grade 4 larynx. How would you proceed?

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<table>
<tr><th>Title</th><td colspan="13">Assessing and managing patients with polytrauma in the perioperative setting</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="13">Specifications:
1. Recognising patients with life-threatening injuries who require immediate stabilisation and resuscitation through a structured tool such as Advanced Trauma and Life Support guidelines
2. Gathering and interpreting information to determine diagnosis and priority goals
3. Initiating and organising resuscitation of unstable patients based on urgency of the situation and resource availability
4. Establishing clear leadership, either by recognising the team leader or assuming the leadership role as appropriate. Delegating task and responsibilities in an appropriate and respectful manner
5. Adapting management plans as the clinical situation evolves and seeking assistance in situations that are complex or new
6. Recognising the potential for pulmonary aspiration during airway management and demonstrating competency in the advanced airway skills, including managing difficult airway and consideration for neck/ spine/ facial/ airway injuries
7. Demonstrating competency in haemodynamic resuscitation including skills in obtaining central venous access
8. Demonstrating understanding of activation of massive blood transfusion protocol
9. Formulating and executing an anaesthesia plan including monitoring, invasive lines, and postoperative disposition, that considers the complexity of the patient’s medical condition (e.g., head injuries)
10. Communicating and coordinating patient care involving other professional disciplines
11. Organising the transfer of patient within the hospital or between hospitals with proper documentation and handover of care
12. Seeking and interpreting multiple sources of information to modify patient care for better outcomes
13. Performing team debrief</td></tr>
<tr><td colspan="13">Limitations:
This EPA excludes the following:
1. Obstetric patients
2. Patients aged less than 16 years old</td></tr>
<tr><th>Sub competencies relevant for this EPA</th><td></td><td>PC1</td><td>✓</td><td></td><td>PC7</td><td>✓</td><td></td><td>SBP1</td><td>✓</td><td></td><td>P1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC2</td><td>✓</td><td></td><td>PC8</td><td></td><td></td><td>SBP2</td><td>✓</td><td></td><td>P2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC3</td><td>✓</td><td></td><td>PC9</td><td>✓</td><td></td><td>SBP3</td><td></td><td></td><td>P3</td><td></td><td></td></tr>
<tr><td></td><td>PC4</td><td>✓</td><td></td><td>PC10</td><td></td><td></td><td>PBL1</td><td>✓</td><td></td><td>ICS1</td><td></td><td></td></tr>
<tr><td></td><td>PC5</td><td>✓</td><td></td><td>MK1</td><td>✓</td><td></td><td>PBL2</td><td></td><td></td><td>ICS2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC6</td><td>✓</td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td></td><td></td><td>ICS3</td><td>✓</td><td></td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	1. Knowledge of **Advanced Trauma Life Support** (ATLS), or equivalent protocols including the performance of a primary and secondary survey with consideration for specific patient factors including the elderly, the young, patients with high body mass index (BMI), pregnant patients and patients with concurrent medical conditions 2. Management of specific patient groups including patients with severe head injury, severe burns, mass casualty (including use of biological or chemical or radioactive agents) 3. Management of massive haemorrhage including classes of shock, complications related to massive blood loss including principles of fluid resuscitation, use of blood products and management of coagulopathy, acidosis, and temperature 4. Management of complications of massive blood transfusion including storage lesions, transfusion reactions, coagulopathy, and application of viscoelastic haemostatic assays in management of coagulopathy 5. Role of point of care ultrasound in management of trauma patient 6. Pharmacology of agents used in resuscitation 7. Team management and post-event debrief 8. Safe transfer of patients within and across different institutions
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations including viscoelastic haemostatic assays 2. Procedural skills • Advanced airway skills including familiarity with various airway devices/ front-of-neck access • Management of ventilator including lung protective ventilation strategy • Vascular access including intra-arterial line/central venous line/ intraosseous line • Needle thoracotomy • Point of care ultrasound 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent where possible • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management, team management)

<table>
<tr><td></td><td>A:</td><td>1. Proactive in reflective learning and self- improvement, receptive to feedback
2. Honest, reliable, conscientious, responsible
3. Recognises own limits and escalates to supervisor when appropriate,
4. Provides patient-centric compassionate care, ensuring patient comfort, privacy, and confidentiality
5. Collegial and respectful team player who is willing to help others
6. Demonstrates professional behaviour in complex and stressful situations
7. Maintains duty of care and patient safety while balancing multiple responsibilities</td></tr>
<tr><td></td><td>E:</td><td>1. Completed ATLS or acquisition of equivalent knowledge/ experience
2. Case log of managing patients with polytrauma (at least 4 cases) over 3 years
3. Teaching sessions relevant to polytrauma
4. Minimum 8 case logs by end R5</td></tr>
<tr><td>Potential risks in case of failure</td><td colspan="2">1. Increase in patient dissatisfaction/ morbidity/ mortality
2. Increase in medico-legal liability
3. Increase in healthcare cost</td></tr>
<tr><td rowspan="4">Sources of information to support summative entrustment decisions</td><td>Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed (To consider evaluators and context)</td></tr>
<tr><td>Entrustment-based Discussion (EBD)</td><td>Minimum 3 by end R3 for level 4 supervision</td><td>Physician Faculty co-managing case
To be completed by end R3
1. Management of head/ neck trauma
2. Management of thoracic/ abdominal trauma
3. Management of pelvic/lower limb injury</td></tr>
<tr><td>Multisource 360° Feedback (MSF)</td><td>2</td><td>EOT (emergency OT) setting
Include nurses/ doctors (at least 3) who work with the resident in EOT

Resident to decide on evaluators and match on the spot for evaluation
• 1 in Junior Residency (R1 to R3)
• 1 in Senior Residency (R4 to R5)</td></tr>
<tr><td>Case Logs</td><td>Minimum cumulative numbers
• 4
• 8</td><td>

To be completed by end R3
To be completed by end R5</td></tr>
</table>

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	Case reflections	Minimum 4	Reviewed by programme *To be completed by end R5* 1. Holistic management of polytrauma patient
Which supervision level when?	Level 4 supervision by end R3		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 5

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
EBD	Any trauma case	Management of head / neck trauma	1. How would you exclude cervical spine injury in this patient? 2. What precautions will you take when inducing general anaesthesia in this patient and why? 3. In this patient before induction, what would you specifically look out for in your assessment of this patient? 4. If this patient had a penetrating neck wound, what would you do? 5. How would you prevent secondary brain trauma? What are your haemodynamic goals? 6. What would you do if you noticed that the airway pressure was elevated and there was evidence of subcutaneous emphysema after intubation? 7. In patients on warfarin, what would be required for reversal?
EBD	Any trauma case	Management of thoracic/ abdominal trauma	1. What are the goals of damage control resuscitation? Describe your fluid management. 2. What is your choice of induction method and airway management for this patient and why? 3. If soon after induction, the blood pressure plummets, what are your goals for management? 4. What would you do during the laparotomy if the blood pressure is persistently low despite transfusion of 1 litre of blood and your noradrenaline requirements are creeping up? 5. How would you decide if a chest tube is required pre-induction or post-induction? 6. Explain how you would use the ROTEM results to guide your resuscitation. 7. What is the definition of massive blood transfusion? What are the potential complications arising from massive blood transfusion?

EBD	Any trauma case	Management of pelvic/lower limb injury	1. What are the means for haemorrhagic control pre-op and intra-op? 2. What are the potential complications arising from such injuries? 3. What is damage control surgery? 4. What are the features for compartment syndrome? 5. What would be your pain management strategy?

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<table>
<tr><th>Title</th><td colspan="2">Managing critically ill patients in the intensive care unit (ICU)</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="2">Specifications:
1. Identifying, stabilising, and resuscitating critically ill patients
2. Triaging of patients for admission to ICU
3. Coordinating interdisciplinary management and disposition of critically ill patients to ensure optimal patient outcomes with appropriate use of ICU resources and transitions of care
4. Initiating and managing organ/system support in critically ill patients including
a. Central nervous system – Sedation, Intracranial pressure management
b. Cardiovascular system– Vasoactive medications, Point-Of-Care-Ultrasound (POCUS), FloTrac®, Pulmonary Artery catheter
c. Respiratory system – Mechanical ventilation /Non-invasive ventilation including High Flow Nasal oxygen.
d. Gastrointestinal system – Stress ulcer prophylaxis, Nutrition
e. Renal system – Renal Replacement Therapy
f. Infection control – Rational use of antibiotics
5. Performing critical care procedures such as intra-arterial and central venous cannulation, vascular catheter insertion for dialysis
6. Performing concise handover of care with pertinent issues highlighted
7. Leading a multidisciplinary critical care team in patient care e.g., during resuscitation
8. Communicating difficult news to patient/ family with sensitivity e.g., breaking bad news, poor prognosis, end-of-life discussions
9. Practicing infection control principles i.e., patient isolation/ precautions, personal protective measures</td></tr>
<tr><td colspan="2">Limitations:
This EPA excludes the following:
1. Patients aged less than 16 years’ old
2. Cardiothoracic surgical intensive care unit and coronary care unit patients</td></tr>
<tr><th>Sub competencies relevant for this EPA</th><td colspan="2">
<table>
<tr><td>PC1</td><td></td><td>PC7</td><td>✓</td><td>SBP1</td><td>✓</td><td>P1</td><td>✓</td></tr>
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</table>
</td></tr>
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Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	For details, please refer to MMed Anaes ICM syllabus https://medicine.nus.edu.sg/wp-content/uploads/sites/30/2021/09/mmed_anaes_b_syllabus.pdf (Pg. 13-15) **1. Applied pharmacology of drugs used in critical care** **2. Applied physiology** **3. Equipment for clinical monitoring** **4. General Care** • Indication and goals for admission • Scoring systems to assess severity • Universal precautions and infection control • Nutrition • Analgesia and sedation • Effect of critical illness on the pharmacokinetics and pharmacodynamics of sedative and analgesic agents • Weaning of sedative and analgesics agents • Use of muscle relaxants • Nosocomial infections • Complications of prolonged intensive care • Transfer of critically ill patients • Communication: inter-disciplinary and patient/relatives **5. Sepsis and multi-organ dysfunction** • Sepsis, severe sepsis, and systemic inflammatory response syndrome (SIRS) • Mechanisms of organ dysfunction • Goal directed therapy of sepsis • Antimicrobial agents • Preventing infection in intensive care **6. Acute circulatory failure and cardiovascular disorders** • Shock • Monitoring: Cardiac output, Tissue perfusion, Arterial blood gases • Heart Failure • Pharmacological CVS support • Pulmonary embolic disorders • Cardiac arrest **7. Respiratory failure and intensive care of respiratory disorders** • Respiratory failure • Principles of management of respiratory failure • Conditions that lead to respiratory failure • Complications of intubation and ventilation • Strategies for weaning patients off ventilatory support • Upper airway obstruction • Tracheostomy

8. Renal and fluid and electrolyte disorders

- Acute renal failure in the critically ill patient
- Conditions which can lead to acute renal failure
- Renal replacement therapy
- Fluid and electrolyte, and acid-base disturbances in the critically ill patient
- Postoperative care of the renal transplant recipient

9. Metabolic and endocrine disorders

- Metabolic response to trauma, surgery, and critical illness
- Acute metabolic and endocrine conditions

10. Neurological and neuromuscular disorders

- Determinants, control, and monitoring of:
  a. Intracranial and intraspinal pressure,
  b. Cerebral blood flow,
  c. Spinal cord perfusion.
- Pathophysiology, assessment, and management of specific conditions:
  a. Acute traumatic brain injury,
  b. Raised intracranial pressure
  c. Prolonged seizures,
  d. Acute spinal cord injury
  e. Hemiplegia, paraplegia, quadriplegia.
  f. Delirium
  g. New neurological impairment
  h. Neurological deterioration (due to various vascular causes)
  i. Cerebral vasospasm
  j. Diabetes insipidus,
  k. Cerebral salt wasting
  l. Neurosurgical patients postoperatively,
  m. Encephalitis and meningitis
  n. Persistent vegetative state
- Confirmation of brain death, issues with communication to family and medico-legal considerations
- Management of the brain-dead patient/ potential organ donors

11. Gastrointestinal disorders

- Assessment and management of the following intra-abdominal conditions:
  a. Oesophageal perforation
  b. Gastrointestinal haemorrhage
  c. Acute pancreatitis
  d. Acute and acute on chronic liver failure
  e. Abdominal sepsis.
  f. Ischemic bowel
  g. Bowel perforation

		h. Intestinal obstruction i. Major abdominal trauma j. Post-major abdominal surgery **12. Haematological and oncological disorders** • Coagulopathy due to: a. Trauma b. Medications c. Disseminated intravascular coagulation • Use of anti-coagulants for prevention and management of venous and arterial thrombosis and thromboembolism • Management of anaemia and thrombocytopaenia • Blood products and transfusion reactions **13. Obstetric patient** • Consideration and requirements • Resuscitation • Specific conditions: a. Severe pre-eclampsia and eclampsia b. Post-partum haemorrhage c. Amniotic fluid embolism
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Management of airway • Intravascular access (venous/ arterial/ Vascath) • Insertion of devices within the nasal, oropharyngeal, oesophageal, bladder and gastric cavity for the purpose of monitoring, extraction of gases or bodily fluids and/or infusion of gases or fluids 3. Communication and interpersonal skills • Patient/ family centric communication including breaking bad news, end of life discussions • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)

	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides patient-centric compassionate care, ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Fundamental Critical Care Support (FCCS) Basic Assessment and Support in Intensive Care (BASIC) 2. Residency Core Intensive Care Medicine (ICM) lectures 3. Site specific ICM teaching, Department Continuing Medical Education (e.g., Grand Ward Rounds, Journal Club, Morbidity and Mortality Rounds) 4. Hospital Infection Control Teaching 5. Relevant Simulation Teaching/Exercises
Potential risks in case of failure	1. Increase in patient morbidity/dissatisfaction/ mortality 2. Increase in patient's family's dissatisfaction/ distress 3. Increase in medico-legal liability 4. Increase in healthcare costs 5. Increased risk of delay in care of other patients	

Sources of information to support summative entrustment decisions	**Tools**	**Number to be completed satisfactorily**	**Additional specifications if needed (To consider evaluators and context)**
	Mini-Clinical Examination (Mini-CEX)	Minimum 1 by end R3 for level 4 supervision	Physician Faculty *To be completed by R3* 1. Outreach/Triage admission to ICU
	Direct Observation of Procedural Skills (DOPS)	Minimum 2 reaching minimal score of 5 for overall assessment.	Physician Faculty *To be completed by end R1* 1. Minimum 1 Intra-arterial (IA) line *To be completed by end R3* 2. Minimum 1 Dialysis vascular access catheter insertion
	Entrustment-based Discussion (EBD)	Minimum 3 (2 by end R2 for level 3 supervision 1 by end R3 for level 4 supervision)	Physician Faculty co-managing case *To be completed by end R2* (For level 3 supervision) 1. Post general surgery critically ill patient 2. Neurosurgical critically ill patient *To be completed by end R3* (For level 4 supervision) 3. Critically ill patient with severe medical comorbidities

	Competency Evaluation	3	Physician Faculty *To be completed by end of rotation* • 2 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
	Multisource 360° Feedback (MSF)	3 In ICU setting	Minimum 1 Faculty, 1 peer, 2 nurses • 2 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
Which supervision level when?	Level 3 supervision by end R2 Level 4 supervision by end R3		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 6

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
MiniCEX	Triage	1. Able to appropriately triage and devise a management plan for patients referred to ICU team 2. Able to effectively communicate with patients and family members in the following challenging scenarios: a. Critically ill patients not suited for ICU admission b. Critically ill patients requiring ICU admission 3. Critically ill patients where further advanced life support measures are deemed futile, and extent of care needs to be delineated/established	a. You are asked to review a 65-year-old female in the ED, who p/w RHC pain and vomiting. She has history of ischemic heart disease and AF. She is jaundiced, mildly confused, and has RUQ tenderness. She was given 4L 0.9% NaCl due to hypotension on presentation. Her current vitals: Temp 39 deg C, RR 30/min, SpO2 92% on NRM, HR 120/min AF, BP 88/48 mmHg Abdo ultrasound shows dilated CBD and enlarged gall bladder with thickened walls. How would you manage this patient? • *ABC, Surviving Sepsis guidelines, broad spectrum antibiotics, source control with decompression and drainage of biliary tract.* b. A 56-year-old woman develops a spontaneous SAH in the ward. She has a GCS of 12 and requires transfer to a tertiary centre for neuro-intervention. Outline the clinical and organisational issues that need to be addressed pre-transfer. • *Coordination and communication, preparation of patient, monitoring, equipment and drugs, personnel* c. Outline your principles of management in the transport of the critically ill patient. • *Justification for transfer, stability for transfer, equipment, monitoring, patient preparation,*

			communication, trained personnel d. You are called to see a 69-year-old male in the general surgical ward who has become profoundly hypotensive (75/40 mmHg). He is POD 5 after palliative surgery for a perforated malignant gastric ulcer. He is barely rousable and SpO2 is 85% on facemask 10L/min. Discuss your plan of management. • *Information regarding his prognosis and expressed wishes, "comfort measures" should only be introduced when considered appropriate (with input from family and primary team), treat underlying cause of deterioration,* e. As part of ICU Outreach and stepdown review, you are seeing a 72-year-old lady with history of severe COPD who was just discharged from the ICU yesterday. She was admitted for cholangitis and decompensated type 2 respiratory failure and had 3 weeks of difficult ICU management and prolonged ventilator weaning process. According to her daughter, the patient lives independently but is limited by severe exertional dyspnoea. What are the principles of managing her should she deteriorate and require admission to ICU? • *Pillars of ethics: beneficence, non-maleficence, justice, autonomy, futility/utility.* • *Prepare for ACP discussion esp. with regards to patient's desire for intubation, if necessary, in the*

			future, the use of NIV and other invasive organ support etc. • *Decision can be made to limit or withdraw therapy if therapy: (1) is unlikely to succeed (physiologically non-plausible), (2) is known to fail in vast majority of times under such conditions and/or associated with adverse consequences, (3) will lead to a QOL that is unacceptable, (4) lead to preservation of state that is unacceptable for the patient* f. That night, the patient develops mucus plugging and becomes drowsy. Her daughter demands for ICU admission. What will you do? • *Depends on results of fam conf and any plans reached.* • *Assess appropriateness and other alternatives.* • *Treat reversible causes – chest physio, nebs saline, supplemental oxygen etc.* *Align with patient's wishes to preserve autonomy*
DOPS	Arterial line	1. Knows appropriate indication/ contraindication for arterial line insertion. 2. Knows the appropriate set up and components of the measurement system 3. Able to interpret information provided by the arterial tracing 4. Troubleshoot commonly encountered device-related complications including site bleeding, hematoma, and difficult insertion etc.	1. What are some indications of arterial line insertion? 2. What are the components of the arterial BP monitoring system? (IA catheter, fluid filled tubing, fluid in tubing, counter pressure fluid bag, pressure transducer, signal conditioning and monitoring software) 3. What information can be gained from an arterial line trace? • Measurements: HR, systolic pressure, diastolic pressure (coronary filling), MAP (systemic

		5. Appropriate consent taking- explains indication, risks, benefits 6. Correct patient, indication, checks for no contraindications 7. Communicates with patient during procedure to ensure comfort, uses LA 8. Appropriate checking of equipment including ultrasound machine if indicated and zeroing transducer 9. Appropriate aseptic technique 10. Proper technique, feels for pulse before attempt 11. Talks to patient throughout procedure 12. Communicates clearly with healthcare team 13. Escalates appropriately to senior when unable to handle 14. Institutes appropriate patient monitoring 15. Demonstrates situational awareness during procedure 16. Manages and disposes sharps safely 17. Appropriate securement of IA line 18. Documents procedure properly	perfusion), pulse pressure, pulse pressure variation, slope of anacrotic limb (associated with AS) • Waveform shape: Slope of anacrotic limb represents aortic valve and LVOT flow, slurred wave in AS, collapsing wave in AR, low dicrotic notch in states with poor peripheral resistance, position, and quality of dicrotic notch as a reflection of the damping coefficient) 4. How would you assess damping/ dynamic response in the arterial pressure transducing system? What do you understand by a square wave test? 5. How would the arterial waveform differ at the various sites of insertion? *Peripheral arterial waveforms have a higher systolic BP, steeper systolic upstroke, lower diastolic BP, lower and later dicrotic notch and wider pulse pressure compared with measurements obtained at the aortic root.*
DOPS	Dialysis catheter	1. Knows appropriate indication/ contraindication for dialysis catheter insertion. 2. Able to justify the choice of site and length of catheter used 3. Troubleshoot commonly encountered device-related complications including site bleeding, hematoma, inadvertent arterial	1. Outline the anatomical structures relevant to the insertion of a femoral dialysis catheter. 2. How would you differentiate the vein from the artery on ultrasound? 3. How would you choose the site of insertion and length of dialysis catheter to be used? 4. What are some of the complications associated with dialysis catheter

		cannulation, and poor lumen flows etc. 4. Appropriate consent taking – explains indication, risks, benefits 5. Ensures no contraindication to dialysis catheter placement 6. Communicates with patient during procedure to ensure comfort, uses LA 7. Appropriate monitoring 8. Appropriate checking of equipment, including ultrasound machine and transducer 9. Strict aseptic technique 10. Knows landmark technique in addition to using ultrasound guided placement 11. Proper technique including pre-procedure scan to identify sonoanatomy 12. Checks the guidewire position on ultrasound after insertion by scanning in transverse and longitudinal planes 13. Ensures dilation of track does not result in guidewire kinks 14. Manages and disposes sharps safely 15. Appropriate securing of dialysis catheter with 3-point suturing 16. Ensures all lumens are freely patent, heparin locked and capped 17. Documents procedure properly 18. Orders and checks chest x ray post line insertion 19. Communicates clearly with healthcare team members	insertion? How can you minimise these complications? 5. At the end of insertion of the dialysis catheter and while flushing it, you find that there is resistance in aspirating and flushing one of the ports. What do you think are some possible reasons and what would you do? 6. You had just inserted a dialysis catheter for a patient. However, on clearing the procedure trolley, you realised that the guidewire had not been removed. How would you manage this situation? 7. After you had dilated and inserted the dialysis catheter into the vessel, bright red blood backflows out from the lumens. Describe your immediate next steps in management. 8. 3 hours into initiation of CRRT, the nurse informs you that access pressures are high. How would you troubleshoot this? 9. A repeat episode of high access pressure occurs despite your initial management steps. The nurse then proceeds to run the CRRT on reverse flow. What do you understand by “reverse flow” and how would it impact dialysis? 10. The dressing around on the dialysis catheter is heavily soaked. On inspection of the insertion site after removal of the dressing, there is oozing around the puncture site. How would you manage this?

		20. Escalates to senior appropriately when unable to handle 21. Demonstrates situational awareness during procedure	
EBD	Post general surgery critically ill patient	1. Understand the basic principles and areas of postoperative care for a post-operative patient admitted to the ICU 2. Management of common post-surgical complications e.g., AKI, ileus, high drain/NG output etc. 3. Checklist for extubation 4. Know the indications and contraindications for commencing enteral or parenteral, and associated complications	1. A 60-year-old woman has a right hemi-hepatectomy for invasive cholangiocarcinoma. She has been admitted to the ICU for post-operative care. What are the problems she may develop in the first 48 hours and how would you treat them? • *Related to major abdominal surgery + (2) specific for hemi-hepatectomy* • *Respiratory, CVS, bleeding, GI failure, liver failure, renal – HRS, ATN, CNS – analgesia, delirium, hepatic encephalopathy, metabolic – high lactate low phosphate, hypoglycaemia, hypercatabolic state, coagulopathy* 2. What are the indications for TPN post major abdominal surgery? 3. Describe how you would order TPN. 4. A 67-year-old man is admitted to the ICU following laparotomy for faecal peritonitis. He developed multi organ failure over the next 2 days, requiring increasing ventilator and inotropic support. He is oliguric, acidotic and has increasing serum creatinine. What are the likely mechanisms for this patient's renal failure? When would you start dialysis for this patient? 5. How would you assess fluid balance in the critically ill patient?

			• *Clinical, fluid balance chart, daily weights, CVP. TTE, IVC assessments, SVV/PPV, CO monitor, bioreactance technology* 6. How would you assess readiness for extubation? What is a spontaneous breathing trial (SBT)? What are some ways you can perform an SBT? Over what duration should it occur? What do you understand by the Rapid Shallow Breathing Index (RSBI)?
EBD	Neurosurgical critically ill patient	1. Recognition of the clinical signs, and basic management of raised ICP 2. Acute management of seizures 3. Prevention, surveillance, and treatment of vasospasm in SAH 4. Goals of management of TBI 5. Able to recognise clinical features of and work up for SIADH, diabetes insipidus, cerebral salt wasting 6. Brain death certification and ICU management of the brain-dead organ donor	1. A 52-year-old patient is admitted to NICU for WFNS IV SAH following an aneurysmal rupture. On day 4 of admission, you notice that the patient has become hyponatraemic (Na 126 mmol/L). What are your differential diagnoses of the acute hyponatremia? Outline your principles of management for his hyponatremia and specific management based on the most likely diagnoses. • *Ddx: SIADH, CSW, hypovolemic hypoNA, hypoadrenalism, hypothyroidism, hypervolemic hyponatremia, inappropriate IV fluids* • *SIADH most likely > CSW* • *Tx: Hypertonic saline, aim > 135. Cannot fluid restrict as she is at risk of vasospasm. Check RP q4-6h. Avoid hypotonic fluids. Check urine osmolality to help guide fluid administration.* 2. You are called to review a 55-year-old female following emergency clipping of a ruptured left MCA aneurysm

			which was completed 3 hours ago. She is intubated and ventilated, with an EVD in situ. She developed frank blood in the EVD and her ICP increased to 57 cmH2). What would be your approach to her initial management? 3. A 45-year-old male is admitted to the ICU following severe traumatic brain injury. He is sedated, intubated, and ventilated. • What are your goals of management with respect to the TBI? • What is "secondary brain injury"? • What are the steps you would take to avoid it? This patient had an ICP monitor inserted. The initial ICP is 32 mmHg. How would you manage the raised ICP? 4. Can you outline the ICU management of a 35-year-old patient who has fulfilled brain death criteria and is awaiting surgery for organ donation?
EBD	Critically ill patient with severe medical comorbidities	1. Able to devise a reasonable management plan for a critically ill patient, taking into account severe medical comorbidities. 2. Interpretation of cardiac output monitors to guide hemodynamic management 3. Theoretical and practical knowledge on basic TTE views and how to diagnose the different types of shock (hypovolemic, obstructive, cardiogenic) 4. Strategies for management of ARDS, including advanced	1. A 55-year-old female is admitted to your ICU with severe respiratory failure caused by a community acquired pneumonia. She has a history of rheumatoid arthritis. What are some factors related to her rheumatoid arthritis that would impact your ICU management? 2. A 35-year-old female is brought into ED with severe acute asthmatic exacerbation. Despite continuous nebulised ventolin, she goes into respiratory failure and is intubated. • Describe your initial ventilator settings and explain the rationale for each of your choices.

		ventilation modes and PEEP titration 5. Understand the basic components in kidney replacement therapy prescription and how to vary them to achieve a clinical target 6. Display reasonable clinical judgment on transfusion triggers for the various cell lines, taking into account patient background and current medical state 7. Management of common medical emergencies e.g., status asthmaticus, status epilepticus and thyroid storm etc.	1 hour later the patient becomes hypotensive. • What are your differential diagnoses? • What do you understand by dynamic hyperinflation? • What are your next steps in management? 3. What are some of the possible approaches to setting PEEP in a ventilated patient with ARDS? • *FiO2/PEEP titration table, use of P-V curve, adjust to maximum static compliance, staircase recruitment maneuverer best PEEP 2.5cmH2O above derecruitment point, transpulmonary pressure* 4. A 32-year-old female is admitted to your ICU for first onset seizures. She had been given doses of IV midazolam and loaded with IV levetiracetam. Despite this she continued to have seizures with persistently depressed GCS. She was intubated using propofol and rocuronium. Currently in the ICU, she is on propofol infusion 100mg/h but continues to have intermittent seizures. Briefly outline your specific management with respect to the seizures. 5. Briefly outline how you would manage a thyrotoxic crisis in the ICU.

Anaesthesiology EPA 7

<table>
<tr><th>Title</th><td colspan="12">Providing anaesthetic care for obstetric patients</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="12">This EPA includes the following nested EPAs:
a. Managing parturient for labour epidural/ combined spinal epidural (CSE)
b. Managing parturient for Lower Segment Caesarean Section (LSCS)
Specifications:
1. Assessing obstetric patients through history taking, physical examination and relevant investigations
2. Locating best available evidence and using it to formulate and justify anaesthetic plans for obstetric patients with consideration for common issues that impact obstetric anaesthesia care, complex issues in critically ill obstetric patients and patient preference
3. Anticipating potential obstetric, medical, or anaesthetic problems and formulating contingency plans
4. Providing patient consultation, counselling, and obtaining informed consent for procedures
5. Making appropriate referrals to other specialties
6. Communicating with patients and their families, and engaging them in shared decision making
7. Coordinating care of patients through effective communication between healthcare professionals in a multidisciplinary team to ensure safe transition of care and handovers especially in complex clinical situations like non obstetric surgery which requires extra perioperative precautions and monitoring for foetal wellbeing
8. Managing the difficult obstetric airway in accordance with the difficult obstetric airway algorithm
9. Performing central neuraxial regional anaesthesia (RA) in the obstetric patient
10. Managing complex obstetric conditions according to the latest guidelines such as severe pre-eclampsia, massive obstetric haemorrhage
11. Organising and managing resuscitation in obstetric emergencies such as eclampsia and massive obstetric haemorrhage, as part of a multidisciplinary team</td></tr>
<tr><td colspan="12">Limitations:
This EPA excludes the following:
1. Obstetric patients requiring intensive care
2. Medically complex mother-foetal dyad presenting for foetal surgery
3. Obstetric patients undergoing complicated non obstetric surgeries e.g., complicated oncological surgeries</td></tr>
<tr><th>Sub competencies relevant for this EPA</th><td></td><td>PC1</td><td>✓</td><td></td><td>PC7</td><td>✓</td><td></td><td>SBP1</td><td></td><td></td><td>P1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC2</td><td>✓</td><td></td><td>PC8</td><td>✓</td><td></td><td>SBP2</td><td>✓</td><td></td><td>P2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC3</td><td>✓</td><td></td><td>PC9</td><td></td><td></td><td>SBP3</td><td></td><td></td><td>P3</td><td></td><td></td></tr>
<tr><td></td><td>PC4</td><td>✓</td><td></td><td>PC10</td><td>✓</td><td></td><td>PBL1</td><td>✓</td><td></td><td>ICS1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC5</td><td>✓</td><td></td><td>MK1</td><td>✓</td><td></td><td>PBL2</td><td>✓</td><td></td><td>ICS2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC6</td><td></td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td></td><td></td><td>ICS3</td><td>✓</td><td></td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	1. Physiological changes in pregnancy including haemodynamic consequences of pregnancy, labour, and delivery 2. Foetal/neonatal/ placental physiology 3. Applied pharmacology relevant to obstetric patients including local anaesthetics, agents affecting uterine tone, opioids and drug interactions, and potential effects on foetus including teratogenic effects 4. Applied anatomy relevant to regional anaesthesia for obstetrics including central neuraxial blocks and pain pathways involved in the first and second stages of labour 5. Pathophysiology of various common complications of pregnancy including gestational diabetes and pre-eclampsia 6. Anaesthetic implications and obstetric management of the following high-risk pregnancy conditions a. Hypertensive disorders of pregnancy – preeclampsia, eclampsia b. Multiple gestation c. Preterm labour including use of tocolytics, dexamethasone d. Abnormal foetal presentations e. Peripartum haemorrhage f. Maternal or foetal infection – HIV, sepsis, chorioamnionitis g. Endocrine disease – diabetes mellitus, thyroid disease h. Substance abuse i. Respiratory diseases – asthma j. Cardiovascular diseases – cardiomyopathy, congenital heart disease, valvular heart disease k. Haematological or thromboembolic diseases – anaemia's, idiopathic thrombocytopenic purpura, von Willebrand's disease l. Morbid obesity m. Musculoskeletal disorders – scoliosis, prior back surgery 7. Methods of providing analgesia/ anaesthesia for routine labour and vaginal delivery, and instrumental delivery such as vacuum extraction and trial of forceps (including drug dosages) 8. Methods of providing anaesthesia for elective or emergency surgical delivery such as use of existing labour epidural, spinal anaesthesia, and general anaesthesia (GA) (including drug dosages) 9. Methods of providing analgesia/ anaesthesia for Examination under anaesthesia (EUA) for postpartum haemorrhage, manual removal of placenta and tubal ligation 10. Anaesthetic considerations for the mother, foetus, and medically complex neonate 11. Resuscitation/ Trauma management of the pregnant patient 12. Resuscitation of the new-born including APGAR scoring, umbilical cord blood gas measurements

	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • RA including spinal, epidural, and combined spinal epidural (CSE) • Obstetric airway skills 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent from anxious patients, patients in labour or teenage mothers, and comforting patients and family in the setting of poor foetal outcome/demise • Healthcare professional/interdisciplinary centric communication to coordinate multidisciplinary care of obstetric patients including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Clinical rotations in obstetrics in senior and junior residency: minimum 4 months in Junior Residency (R1 to R3), minimum 1 month in Senior Residency (R4 to R5) 2. Cases logs 3. Interprofessional team training workshop/simulation including maternal and neonatal resuscitation
Potential risks in case of failure	1. Increase in maternal morbidity/ dissatisfaction/mortality 2. Increase in adverse foetal/neonatal outcomes 3. Increase in medico-legal liability 4. Increase healthcare cost	

Sources of information to support summative entrustment decisions	**Tools**	**Number to be completed satisfactorily**	**Additional specifications if needed (To consider evaluators and context)**
	Mini-Clinical Examination (Mini-CEX)	Minimum 1 by end R3 for level 3 supervision	Physician Faculty *Please refer EPA 7b*
	Direct Observation of Procedural Skills (DOPS)	Minimum 3 reaching minimal score of 5 for overall assessment.	Physician Faculty *Please refer EPA 7a*

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	Entrustment-based Discussion (EBD)	Minimum 3 (1 by end R3 for level 3 supervision 2 by end R5 for level 4 supervision)	Physician Faculty co-managing case *Please refer EPA 7a, 7b*
	Competency Evaluation	Minimum 2	• 1 in Junior Residency (R1 to R3) rotation • 1 in Senior Residency (R4 to R5) rotation
	Multisource 360° Feedback (MSF)	2	Each evaluation comprises the following evaluators: physician faculty, nurses, and peers (minimum 6 evaluators in total) • 1 in Junior Residency (R1 to R3) rotation • 1 in Senior Residency (R4 to R5) rotation
	Case Logs	Minimum cumulative numbers: • 40 Obstetric RA • 20 Lower Segment Caesarean Section	Reviewed by programme *To be completed by end of R3* **Minimum 10 directly observed cases** for **level 3 supervision** • Includes epidural, spinal, CSE • Includes LSCS done under RA, GA, and high-risk cases *To be completed by end of R5* For level 4 supervision Minimum numbers • 40 Obstetric RA • 20 Lower Segment Caesarean Section
	Case reflections	Minimum numbers: • 5 Lower Segment Caesarean sections • 5 Labour Analgesia	Reviewed by programme *To be completed by end of R5* • Includes management of pre-eclamptic patients • patients with placenta praevia majors and cardiac disease in pregnancy
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?	1 year	

Anaesthesiology EPA 7a

<table>
<tr><td>Title</td><td colspan="12">Managing parturient for labour epidural/ combined spinal epidural (CSE)</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td colspan="12">Specifications:
1. Identifying high risk patients, complex ethical situations, and those with contraindications to labour regional anaesthesia (RA) and escalating situation to supervisor
2. Counselling patient for labour analgesia; eliciting patient’s values, goals, and preferences; discusses risks and benefits of RA and/or alternatives
3. Formulating and implementing an appropriate labour epidural analgesia plan
4. Performing central neuraxial anaesthesia including using ultrasound to identify interspinous spaces and the midline in patients with complex neuraxial anatomy
5. Managing failed/ inadequate labour epidural analgesia and complications related to labour epidural/ CSE such as Post Dural Puncture Headache (PDPH), hypotension, backache, dural puncture, foetal bradycardia after initiation of epidural/ CSE</td></tr>
<tr><td colspan="12">Limitations:
1. Refer main EPA 7</td></tr>
<tr><td>Sub competencies relevant for this EPA</td><td colspan="12"><table>
<tr><td>PC1</td><td>✓</td><td>PC7</td><td>✓</td><td>SBP1</td><td>✓</td><td>P1</td><td>✓</td></tr>
<tr><td>PC2</td><td></td><td>PC8</td><td></td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
<tr><td>PC3</td><td>✓</td><td>PC9</td><td></td><td>SBP3</td><td></td><td>P3</td><td></td></tr>
<tr><td>PC4</td><td></td><td>PC10</td><td>✓</td><td>PBL1</td><td></td><td>ICS1</td><td>✓</td></tr>
<tr><td>PC5</td><td></td><td>MK1</td><td>✓</td><td>PBL2</td><td></td><td>ICS2</td><td>✓</td></tr>
<tr><td>PC6</td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
</table></td></tr>
</table>

Anaesthesiology EPA 7a *(Click here to return to Table of Content)*

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	1. Applied physiology, mechanism and stages of labour and effect on maternal hydration, ventilation and haemodynamic 2. Anatomy and sonoanatomy relevant to central neuraxial RA 3. Pain pathways for first and second stages of labour 4. Mechanism of action of central neuraxial blockade in labour 5. Pharmacology of drugs used for central neuraxial blockade in labour 6. RA techniques (including drug dosages, limitations, and complications) used to provide analgesic / anaesthetics for: • Routine labour and vaginal delivery • Instrumental delivery – vacuum extraction, trial of forceps 7. Management of • Foetal distress/non-reassuring foetal status • Hypotension • High/total spinal block • Inadvertent dural puncture • Inadequate/failed central neuraxial block including considering differential diagnosis for persistent pain such as uterine rupture • Systemic local anaesthetic toxicity • Postpartum neurological deficits including assessment, differential diagnosis, appropriate referrals and follow up
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • RA including spinal, epidural, and combined spinal epidural 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent in anxious obstetric patients in labour, teenage mothers • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of obstetric patients including accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)
	A:	As in main EPA 7
	E:	1. Clinical rotations: minimum 4 months as Junior Residency (R1 to R3), minimum 1 month in Senior Residency (R4 to R5) 2. Case Logs: Minimum requirements: 40 RA (epidural, spinal, CSE) 3. Teaching sessions on labour epidural/ CSE

Potential risks in case of failure	As in main EPA 7		
Sources of information to support summative entrustment decisions	**Tools**	**Number to be completed satisfactorily**	**Additional specifications if needed (To consider evaluators and context)**
	Direct Observation of Procedural Skills (DOPS)	Minimum 3 by end R3 reaching minimal score of 5 for overall assessment.	Physician Faculty *To be completed by end R3* 1. Labour epidural / CSE
	Entrustment-based Discussion (EBD)	Minimum 2 (1 by end R3 for level 3 supervision 1 by end R5 for level 4 supervision)	Physician Faculty co-managing case *To be completed by end R3* Minimum 1 For level 3 supervision 1. Troubleshooting inadequate or failed labour epidural 2. Foetal distress/non-reassuring foetal status post RA 3. Hypotension 4. High/total spinal block 5. Inadvertent dural puncture *To be completed by end R5* Minimum 1 For level 4 supervision 1. Managing complications of labour epidural/ CSE like postpartum neurological deficit, post dural puncture headache
	Multisource 360° Feedback (MSF)	Refer main EPA 7	Refer main EPA 7
	Case Logs	Minimum cumulative numbers: 40	*To be completed by end R3* For level 3 supervision RA (epidural, spinal, CSE) 1. 10 direct supervision of labour epidural For level 4 supervision 2. Minimum 40

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	Case reflections	Minimum 5	Reviewed by programme *To be completed by end R5* 1. Challenging labour epidural/ CSE
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 7a

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
DOPS	Labour combined spinal epidural (CSE)/epidural	Troubleshooting inadequate or failed labour epidural Foetal distress/non-reassuring foetal status post RA Hypotension High/total spinal block Inadvertent dural puncture Click on this link for a detailed guide to EBD and expectations: EBD_EPA7A_JR.docx	1. What would you do in the initial management of the breakthrough pain? 2. If soon after the CSE is performed, the CTG shows a deceleration to 100/min, what would you do? 3. How would you recognise and manage a high spinal?
EBD	Labour combined spinal epidural (CSE)/epidural Maybe done with one of the DOPS for labour epidural	Troubleshooting inadequate or failed labour epidural Foetal distress/non-reassuring foetal status post RA Hypotension High/total spinal block Inadvertent dural puncture Click on this link for a detailed guide to EBD and expectations: EBD_EPA7A_JR.docx	1. A term nulliparous woman presented at 3–4 cm cervical dilation in labour with irregular contractions. A combined spinal epidural was established with intrathecal administration of 2mg ropivacaine 0.1% and fentanyl 15 mcg diluted with saline to a total of 2 mls and maintenance using patient-controlled epidural analgesia (PCEA) with low-dose mixture ropivacaine 0.1% + fentanyl 2 mcg/ml of background infusion of 8-mls/h and 5-mls boluses available every 15 minutes as required. Four hours later, the patient complained of numbness on her left side and increasing discomfort on her right side. On physical examination, she was unable to move her left leg but had complete range of motion of her right leg. 2. You are administering an epidural for labour analgesia in a 25-year-old

			primigravida, with a 16G Tuohy needle when a wet tap occurs For more entrustment questions please click below: EBD_EPA7A_JR.docx
EBD	Labour combined spinal epidural (CSE)/epidural *(To be completed before solo call as SR)*	Managing complications of labour epidural/ CSE like postpartum neurological deficit, post dural puncture headache Click below for a detailed guide to EBD and expectations: EBD_EPA7A_SR.docx	A previously healthy 38-year-old gravida 1, para 0 was admitted to labour and delivery in active labour at 38 + 5 weeks estimated gestational age. The patient received epidural anaesthesia for pain control. She progressed to complete dilation and effacement and began pushing efforts at +1 station. During the second stage of labour, different positioning techniques were utilised while pushing. No further descent of the foetal head was made over the course of four hours despite adequate contractions and active pushing, and she underwent an uncomplicated primary low transverse caesarean section for arrest of descent. At 24 hours postoperatively, the patient complained of right-sided numbness over her lateral calf and the dorsum of her foot. She was able to stand and walk, but not without dragging her right foot and toes. An examination revealed preserved bilateral hip, knee, and ankle reflexes, with weak right-sided ankle dorsiflexion and plantar flexion, manifesting as a foot drop. The patient also reported difficulty passing urine, which required intermittent catheterisation. Magnetic resonance imaging of the lumbar spine was obtained to evaluate for epidural hematoma, which showed no remarkable findings.

			Click below for a detailed guide to EBD and expectations:  EBD_EPA7A_SR.docx

Anaesthesiology EPA 7b

<table>
<tr><td>Title</td><td>Managing parturient for Lower Segment Caesarean Section (LSCS)</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td>Specifications:
1. Assessing patient through history taking, physical examination and relevant investigations
2. Identifying and escalating care of high-risk parturient presenting for LSCS
3. Formulating and implementing an appropriate anaesthetic plan for patients presenting for LSCS
4. Counselling patients for anaesthesia for caesarean section by:
• Eliciting patient’s values, goals, and preferences
• Discussing risks, benefits of anaesthesia and/or alternatives.
5. Managing inadequate/failed spinal anaesthetic and its complications such as high block, hypotension, and neurological deficits
6. Managing high risk obstetric patients requiring lower segment caesarean section, such as those with preeclampsia, using a multidisciplinary approach
7. Recognising crisis situations such as malignant hyperthermia or local anaesthetic systemic toxicity, calling for help early and initiating management
8. Anticipating and managing difficult obstetric airway using standard obstetric airway algorithm
9. Coordinating care of patients in clinical situations utilising the roles of the interprofessional teams
10. Performing safe and effective transition of care/handovers in routine clinical situations.
11. Managing complex or stressful situations such as massive peripartum haemorrhage, complex ethical situations</td></tr>
<tr><td>Limitations:
In addition to those mentioned under limitations of main EPA 7, this EPA excludes the following:
1. Patients with abnormal placentation
2. Ex utero intrapartum treatment (EXIT) procedure</td></tr>
<tr><td>Sub Competencies relevant for this EPA</td><td>
<table>
<tr><td>PC1</td><td>✓</td><td>PC7</td><td>✓</td><td>SBP1</td><td></td><td>P1</td><td>✓</td></tr>
<tr><td>PC2</td><td>✓</td><td>PC8</td><td>✓</td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
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<tr><td>PC6</td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
</table>
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</table>

Anaesthesiology EPA 7b *(Click here to return to Table of Content)*

<table>
<tr><td rowspan="5">Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment</td><td>K:</td><td colspan="3">1. Methods of providing anaesthesia for elective or emergency surgical delivery such as use of existing labour epidural, spinal anaesthesia, and general anaesthesia (GA) (including drug dosages, advantages, and disadvantages)
2. General anaesthesia for LSCS including
• Indications for general anaesthesia
• Precautions for risk of pulmonary aspiration
• Agents used for induction and maintenance of anaesthesia
• Effects of the GA agents on uterine tone and foetal perfusion
• Effects of the general anaesthesia on the foetus
• Effect of induction to delivery interval, and uterine incision to delivery interval
• Anaesthetic depth management and awareness during GA
3. Pharmacology of utero tonics
4. Maternal and foetal anaesthetic risk of urgent or emergent surgical delivery
5. Post LSCS pain management</td></tr>
<tr><td>S:</td><td colspan="3">As in main EPA 7</td></tr>
<tr><td>A:</td><td colspan="3">As in main EPA 7</td></tr>
<tr><td>E:</td><td colspan="3">1. Clinical rotations: minimum 4 months in Junior Residency (R1 to R3), minimum 1 month in Senior Residency (R4 to R5)
2. Case Logs: Minimum requirements 20 LSCS cases including high risk in Junior Residency (R1 to R3)</td></tr>
<tr><td>Potential risks in case of failure</td><td colspan="4">As in main EPA 7</td></tr>
<tr><td rowspan="4">Sources of information to support summative entrustment decisions</td><td colspan="2">Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed (To consider evaluators and context)</td></tr>
<tr><td colspan="2">Mini-Clinical Examination (Mini-CEX)</td><td>Minimum 1 by end R3 for level 3 supervision</td><td>Physician Faculty
To be completed by end R3
1. Lower segment Caesarean section
To include discussion on management of GA for LSCS in pre-eclamptic patient coming for LSCS</td></tr>
<tr><td colspan="2">Entrustment-based Discussion (EBD)</td><td>Minimum 1 by end R5 for level 4 supervision</td><td>Physician Faculty co-managing case
To be completed by end R5
1. To include discussion on management of GA for LSCS in which patient develops obstetric haemorrhage</td></tr>
<tr><td colspan="2">Competency Evaluation</td><td>Refer main EPA 7</td><td>Refer main EPA 7</td></tr>
</table>

Anaesthesiology EPA 7b *(Click here to return to Table of Content)*

	Multisource 360° Feedback (MSF)	Refer main EPA 7	Refer main EPA 7
	Case Logs	Minimum 20	Reviewed by programme *To be completed by end R3* • Includes LSCS done under RA, GA, and high-risk cases
	Case reflections	Minimum 5	Reviewed by programme *To be completed by end R5* • Management of challenging patients e.g., pre-eclamptic patients, patients with placenta praevia majors and cardiac disease in pregnancy
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 7b

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Lower segment Caesarean section	Management of GA for LSCS in pre-eclamptic patient coming for LSCS Click on this link for a detailed guide to EBD and expectations EBD_EPA7B_JR.docx	1. The patient presented with severe pre-eclampsia and her BP on the initial presentation was 200/110mmHg. She has symptoms of headaches and generalised oedema. The obstetric team has started her on MgSO4 infusion after a loading dose and on a labetalol plus hydralazine infusion. 6 hours from initiation of treatment, her BP ranges between 125/90 to 135/95 mmHg. Blood results show: Hb 12, wcc 16, plt 80 INR 1.5, PT 15, APTT 40 Liver function is deranged with elevated aminotransferases and LDH. The obstetrician is keen on an emergency LSCS for severe pre-eclampsia 1. What are some of the pre-operative optimisations you would want before starting this case? 2. What are the key modifications for general anaesthesia in this patient? a. Considerations of blood pressure control in this patient? b. Considerations for fluid management in this patient? c. Considerations for reversal and emergence for this patient? For more entrustment questions please click on this link EBD_EPA7B_JR.docx
EBD	Lower segment Caesarean section	Management of GA for LSCS in which patient develops obstetric haemorrhage	30-year-old primigravida. BMI 20kg/m2, 41+5 weeks induction of labour for postdates. Epidural inserted and working. Labour augmented with oxytocin infusion. 8 hours later failure to progress for LSCS.

	(To be completed before solo call)	Click on this link for a detailed guide to EBD and expectations  EBD_EPA7B_SR.docx	Baseline Hb 10.1, wcc 18, plt 220, T&S valid Epidural top-up of 20mls 2% lignocaine with bicarb and adrenaline + 50mcg Fentanyl was given. Block is working well. Post-delivery IV Carbetocin given with oxytocin infusion 30U @ 125mls/hr. BP suddenly drop from 105/65 to 70/40 with an increase in HR from 105 to 125. 1.2L blood in suction with ongoing blood loss. The patient complained of feeling unwell. The surgeon reports bleeding from a traumatic extension of the uterine incision with some uterine atony 1. How would you prioritise what you would do? 2. Was this obstetric haemorrhage expected? How severe do you think it is? 3. When and how would you convert to general anaesthesia? For more entrustment questions please click on this link  EBD_EPA7B_SR.docx

Anaesthesiology EPA 8

<table>
<tr><th>Title</th><td colspan="5">Providing anaesthetic care for patients in a non-operating theatre setting</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="5">Specifications:
Remote locations for procedures that require Anaesthesia care may include the following settings:
• Diagnostic Radiology and Intervention Radiology Centre
• Cardiac Catheterisation Laboratory
• Radiotherapy Centre
• MRI Suite
• Endoscopy Centre
• Ambulatory Surgical Centre
• Burns Centre
• Dental Centre
Patients who require anaesthesia care in remote locations can range from well patients who need sedation to tolerate a procedure, to patients who require anaesthesia support because they are in a higher risk category (e.g., endovascular clot retrieval for stroke).
1. Evaluating suitability of patients for procedures in remote locations
2. Formulating and executing anaesthesia plans for patients undergoing procedures and interventions in remote locations
3. Coordinating and communicating between multidisciplinary professionals to facilitate patient care</td></tr>
<tr><td colspan="5">Limitations:
This EPA excludes the following:
1. Procedures involving the heart and/or access to heart chambers</td></tr>
<tr><th>Sub-competencies relevant for this EPA</th><td colspan="5"><table>
<tr><td>PC1</td><td>✓</td><td>PC7</td><td>✓</td><td>SBP1</td><td></td><td>P1</td><td>✓</td></tr>
<tr><td>PC2</td><td>✓</td><td>PC8</td><td>✓</td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
<tr><td>PC3</td><td>✓</td><td>PC9</td><td></td><td>SBP3</td><td></td><td>P3</td><td></td></tr>
<tr><td>PC4</td><td>✓</td><td>PC10</td><td></td><td>PBL1</td><td>✓</td><td>ICS1</td><td>✓</td></tr>
<tr><td>PC5</td><td>✓</td><td>MK1</td><td>✓</td><td>PBL2</td><td></td><td>ICS2</td><td>✓</td></tr>
<tr><td>PC6</td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
</table></td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	**1. Diseases or conditions that require interventional procedures in remote locations** • Liver diseases (intrahepatic cysts) and portal hypertension (requiring transjugular intrahepatic portosystemic shunt, percutaneous radiofrequency ablation therapy) • Paroxysmal supraventricular tachycardia (for radiofrequency catheter ablation) • Cerebral aneurysm (for endovascular intervention) • Brain tumour (for radiotherapy) • Biliary duct disease: Choledocholithiasis, biliary duct stenosis (for endoscopic retrograde cholangiopancreatography and interventions) • Renal cyst • Burns injury **2. Interventional procedure specifics including procedure related requirements and complications** • Transjugular intrahepatic portosystemic shunt for portal hypertension • Endoscopic retrograde cholangiopancreatography and interventions • Radiofrequency catheter liver cysts, renal cysts • Cardioversion • Cerebral endovascular interventions (hemodynamic targets) • Physics and physiologic effects of imaging modalities (magnetic resonance imaging (MRI) scan, intensive imaging scan, computerised axial tomography scan), types of radiation, and radiotherapy • Radiocontrast media (barium-based (orally); iodinated agents and gadolinium contrast media (intravenously) **3. Specific features/ hazards of remote locations** • Diagnostic Radiology and Intervention Radiology Centres, Cardiac Catheterisation Laboratories, Radiotherapy Centres • Magnetic resonance imaging suites • Endoscopy Centres • Ambulatory Surgical Centres • Burns Centres • Dental Centres **4. Anaesthesia considerations for procedures in remote locations** **i. Working space** • Central gas supplies • Equipment available for monitoring, drug delivery, oxygen delivery, anaesthesia delivery warming fluid, resuscitation, airway management • Imaging table • Radiation safety • Safety guidelines (in magnetic resonance suite, for procedures requiring image intensification, and radiotherapy)

		• Environmental temperature control • Dental chair anaesthesia • Shared airway (in dental surgery, in prone position) • Skilled assistants **ii. Crisis management**
	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Management of airway • Intravascular access (venous/ arterial/ Vascath) • Insertion of devices within the nasal, oropharyngeal, oesophageal, bladder and gastric cavity for the purpose of monitoring, extraction of gases or bodily fluids and/or infusion of gases or fluids 3. Communication and interpersonal skills • Patient/ family centric communication including breaking bad news, end of life discussions • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides efficient patient-centred compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Clinical rotations involving remote locations 2. Teaching sessions on remote anaesthesia 3. Case logs of cases requiring anaesthetic care in remote location: Senior Residency (R4 to R5)
Potential risks in case of failure	1. Increase in patient morbidity/dissatisfaction/mortality 2. Increase in medico-legal liability 3. Increase in healthcare costs	

Anaesthesiology EPA 8 *(Click here to return to Table of Content)*

	Tools	Number to be completed satisfactorily	Additional specifications if needed (To consider evaluators and context)
Sources of information to support summative entrustment decisions	Entrustment-based Discussion (EBD)	Minimum 2 (At most only 1 paediatric patient may be discussed) 1 by end R3 for level 3 supervision 1 by end R5 for level 4 supervision)	Physician Faculty co-managing case *To be completed by end of R3* Minimum 1 For level 3 supervision • MRI suite *To be completed by end of R5* Minimum 1 For level 4 supervision • Interventional radiology suite
	Case Logs	Minimum cumulative numbers • 5 • 10	Reviewed by programme *To be completed by end of R3* *To be completed by end of R5*
	Case reflections	Minimum numbers • 5	Reviewed by programme *To be completed by end of R5* • Challenges and learning points related to providing care for patients in remote locations
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 8

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
EBD	MRI	1. Hazards in the environment and safety guidelines 2. Anaesthetic options for procedure 3. Crisis management	1. What are the safety concerns with working in the magnetic resonance environment? 2. What safety features are in place 3. What are the concerns with providing anaesthetic care for patients in this environment? 4. If your patient is having a scan and the ECG shows a bizarre tracing, how would you respond? 5. How is the monitoring equipment used in the MRI environment different from those in OT? 6. If your patient developed severe hypotension after administration of gadolinium contrast, and the capnography showed a tracing consistent with bronchospasm with tachycardia, what would you do? 7. What anaesthetic options are available for a patient undergoing a scan for MRI neck? 8. What anaesthetic options are available for a patient undergoing abdominal and pelvic scan for delineation of suspected renal tumour? 9. The radiologist requests for a "breath hold" in an intubated adult patient receiving an MRI scan of the abdomen. How would you perform this? 10. The patient has a sudden cardiac arrest while inside the MRI tunnel. What would be your initial response?
EBD	Interventional radiology suite	1. Hazards in the environment and safety guidelines 2. Anaesthetic options for procedure 3. Crisis management 4. Knows the difference between non- operating room anaesthesia (NORA) and operating room anaesthesia (ORA)	1. What are the safety concerns with working in the interventional radiology suite? 2. Where are the radiation safety equipment (e.g., lead gowns) found in your institution? 3. What investigations would you look out for in this patient knowing that radiocontrast has to be administered?

		• Less trained staff and equipment immediately available. • Often no scavenging/N2O. • No dedicated post-GA monitoring area.	What if the patient has type 2 diabetes and is on metformin? 4. What are the anaesthetic concerns with a 60-year-old lady (background hypertension, hyperlipidaemia and mild Coronary Artery Disease) planned for coiling of a left PCOM artery aneurysm 2 days after a subarachnoid haemorrhage? a. Patient needs to be immobile (GA, intubated) b. Limited access to patient (needs good IV lines and Arterial line monitoring) c. Hemodynamically stable induction d. Avoid further bleeding while maintaining CPP (Target SBP 100-140mmHg) e. Secure airway with ETT f. Invasive monitoring g. Maintain normocapnia and normothermia. The interventional radiologist suddenly reports that the aneurysm has ruptured. What do you do? 5. What are the differences between non-operating room anaesthesia (NORA) and operating room anaesthesia (ORA)?

Anaesthesiology EPA 9

<table>
<tr><th>Title</th><td colspan="13">Providing anaesthetic care for paediatric patients aged between 3 to 16 years old</td></tr>
<tr><th rowspan="2">Specifications and limitations</th><td colspan="13">Specifications:
1. Assessing the child pre-procedure and recognising the need for optimisation of medical condition, relevant preoperative investigation, and premedication for anxiolysis
2. Counselling and obtaining informed consent from legal guardian, and assent from the child as appropriate, for sedation/ general anaesthesia using age- appropriate language
3. Formulating and implementing an anaesthetic plan that considers the child's physiological, psychological, and surgical condition including depth of hypnosis required, mode of induction, fasting duration, venous access, fluid management, airway management, temperature management, pain management, choice of appropriate monitoring, equipment and drugs, and emergence and recovery from general anaesthesia/ sedation
4. Initiating management of common anaesthesia related complications/ crisis such as laryngospasm, and calling for expert help appropriately
5. Assessing and managing acute pain in children and referring complex pain patients to a paediatric pain consultant
6. Providing initial age-appropriate resuscitation in a critically ill child including trauma
7. Providing anaesthetic care for a child in a remote location such as a diagnostic imaging suite
8. Communicating information effectively with relevant health care team members through verbal and accurate documentation</td></tr>
<tr><td colspan="13">Limitations:
This EPA does not include the following:
1. Neonates, infants, and children aged less than 3 years old
2. Children ASA 4 and above
3. Children with congenital heart disease, potentially difficult airway or undergoing airway procedures, one- lung ventilation, neurosurgical emergencies</td></tr>
<tr><th>Sub competencies relevant for this EPA</th><td></td><td>PC1</td><td>✓</td><td></td><td>PC7</td><td>✓</td><td></td><td>SBP1</td><td></td><td></td><td>P1</td><td></td><td></td></tr>
<tr><td></td><td>PC2</td><td>✓</td><td></td><td>PC8</td><td>✓</td><td></td><td>SBP2</td><td>✓</td><td></td><td>P2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC3</td><td>✓</td><td></td><td>PC9</td><td></td><td></td><td>SBP3</td><td></td><td></td><td>P3</td><td></td><td></td></tr>
<tr><td></td><td>PC4</td><td>✓</td><td></td><td>PC10</td><td>✓</td><td></td><td>PBL1</td><td>✓</td><td></td><td>ICS1</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC5</td><td>✓</td><td></td><td>MK1</td><td>✓</td><td></td><td>PBL2</td><td></td><td></td><td>ICS2</td><td>✓</td><td></td></tr>
<tr><td></td><td>PC6</td><td></td><td></td><td>MK2</td><td>✓</td><td></td><td></td><td></td><td></td><td>ICS3</td><td>✓</td><td></td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	**1. Normal anaesthesia relevant anatomy and physiology of neonates, infants, and children** a. **Cardiovascular** i. Transitional circulation ii. Normal parameters across the different age range b. **Respiratory and airway** i. Development, anatomy, surfactant ii. Pulmonary function/lung volumes in children vs adults and implications iii. Airway changes that occur as a child develops from infancy to adulthood iv. Normal respiratory parameters c. **Neurological** i. Normal developmental milestones ii. Assessment of conscious level in children d. **Thermoregulation** i. Thermogenesis and heat loss in children ii. Thermoneutral zone iii. Prevention and management of hypothermia e. **Renal function** development and implications on i. Fluid and electrolyte requirements ii. Drug handling f. **Hepatic function** development and implications on i. Glucose metabolism ii. Drug handling g. **Nutritional requirements** h. **Haematological** i. Physiological anaemia ii. Foetal haemoglobin versus adult haemoglobin iii. Coagulation i. **Immunological development** i. Vaccination 2. **Pharmacological differences from adults** a. Pharmacodynamic and pharmacokinetic profiles b. Drug toxicities preferentially occurring in children c. Effects of anaesthetics on neurodevelopment/ psychology **3. Perioperative Management of neonatal and paediatric patient: Medical issues** a. The premature neonate and problems related to prematurity b. The ex-premature infant c. The sick neonate/ critically ill child/ assessment of the circulatory system: e.g., degree of dehydration

d. The syndromic/ dysmorphic child: e.g., Trisomy 21
e. Paediatric medical problems and implications for anaesthesia
   i. Atopy: Asthma; Allergic rhinitis; Eczema.
   ii. Congenital heart disease
   iii. Obstructive sleep apnoea
   iv. Stridor and causes
   v. Neuromuscular disease: Cerebral palsy; Epilepsy; Muscle disease e.g., Duchenne's/ Becker's muscular dystrophy, myotonias, spinal muscular atrophy etc
   vi. Developmental delays/ autism/ learning difficulties
   vii. Gastrointestinal disease
   viii. Endocrine/metabolic disease
   ix. Oncology: Cancer and treatment possibilities, immuno-compromised state, anterior mediastinal mass
   x. Rheumatological disease
   xi. Haematological disease
   xii. Primary immune deficiency in childhood (SCID severe combined immune deficiency)
   xiii. Skeletal abnormalities

4. Preoperative Issues

a. Medical legal issues relating to paediatric practice in respect of Consent, Assent, Restraint and Research and the concept of 'Gillick competence'
b. Fasting guidelines; Preoperative instructions and Premedication
c. Child with the potentially full stomach: Rapid sequence induction (modified)
d. Management of patients, parents, and carers in patients with cognitive, communication or behavioural problems

5. General Intraoperative Issues

a. Anaesthetic Agents and Techniques including induction agents/ techniques for general anaesthesia and regional anaesthesia
b. Fluid therapy and blood replacement
c. Management of difficult paediatric airway
d. Recognition and management of airway obstruction in children

6. **Equipment for neonatal/paediatrics patients**

a. Breathing circuits
b. Airway equipment
c. Mechanical ventilation
d. Thermal Control

7. Paediatric Resuscitation

a. Basic cardiopulmonary life support
b. Specific management

		i. Cardiac arrest ii. Respiratory arrest iii. Laryngospasm iv. Bronchospasm v. Aspiration of gastric contents vi. Tension pneumothorax vii. Shock viii. Anaphylaxis ix. Latex allergy x. Sepsis xi. Post-tonsillectomy haemorrhage xii. Gas embolism xiii. Fat embolism xiv. Raised intracranial pressure xv. Local anaesthetic systemic toxicity xvi. Malignant hyperthermia xvii. Coagulopathy xviii. Life threatening glucose, electrolyte, or acid-base disturbances xix. Electrocution xx. Poisoning xxi. Burns 8. **Post-operative Issues and management** a. Postoperative agitation b. Postoperative Nausea and Vomiting 9. **Paediatric Pain management** a. Different pain assessment tools b. Prescription of effective analgesia appropriate for individual's pain intensity, pattern, and clinical condition c. Anticipation and management of side effects/ complications related to analgesia d. Identification of neuropathic/ complex pain requiring specialist input e. Appreciation of the role of non- pharmacological methods in pain management **10. Neonatal Surgical conditions and anaesthetic management** a. Pyloric stenosis b. Meningomyelocele c. Tracheoesophageal fistula d. Omphalocele/ gastroschisis e. Congenital diaphragmatic hernia f. Intestinal obstruction g. Necrotising enterocolitis **11. Paediatric sedation** **12. Non-operating room anaesthesia for paediatric patients** Indications/contraindications; patient selection; anaesthesia considerations and techniques

	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Management of paediatric airway • Intravascular access either direct or ultrasound guided • Simple regional anaesthetic blocks like penile and ilioinguinal nerve blocks 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent from parent/ legal guardian and using age-appropriate language for the child • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management, time management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides efficient patient-centred compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Clinical rotation: min 4 months in JR (R1 to R3) and 2 months in SR (R4 to R5) 2. Case Logs 3. Simulation 4. Teaching sessions/ workshops/ webinars
Potential risks in case of failure?	1. Increase in patient morbidity and mortality. 2. Increase in medico-legal liability. 3. Increased patient/family dissatisfaction 4. Increase in healthcare costs	

Anaesthesiology EPA 9 *(Click here to return to Table of Content)*

	Tools	Number to be completed satisfactorily	Additional specifications if needed (To consider evaluators and context)
Sources of information to support summative entrustment decisions	Mini-Clinical Examination (Mini-CEX)	Minimum 2 by end R3 for level 3 supervision	Physician Faculty *To be completed by end R2* For level 3 supervision Minimum 1 • GA/ SR case, ASA 1-2 aged 3-10 years *To be completed by end R3* Minimum 1 in R3 • GA/IPPV case, ASA 2-3, age 3-5 years
	Entrustment-based Discussion (EBD)	Minimum 2 (1 by end R3 for level 3 supervision 1 by end R5 for level 4 supervision)	Physician Faculty co-managing case *To be completed by end R3* For level 3 supervision Minimum 1 • Paediatric trauma patient requiring surgery *To be completed by end R5* Minimum 1 For level 4 supervision • Special needs child requiring anaesthesia
	Competency Evaluation	3	• 2 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
	Multisource 360° Feedback (MSF)	2	Minimally 1 peer, 2 faculty, 2 nurses • 1 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
	Case Logs	Junior Residency (R1 to R3) Minimum 125	Minimum case numbers *To be completed by end R3* 125 cases: • < 3 months 5 cases • < 3 years 20 cases • < 12 years 100 cases
		Senior Residency (R4 to R5) Minimum 30	Minimum case numbers To be completed by end R5
Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 9

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	GA/ SR case, ASA 1-2 aged 3-10 years	1. Counselling for informed consent 2. Management of laryngospasm 3. Pain Management options 4. Management of postoperative agitation	1. If this patient had URTI, how would you counsel the parent? 2. If after removal of the LMA deep and transferring the patient to the trolley, the oxygen saturation starts to fall to 95%, what are your differential diagnoses and what would you do? 3. If the child starts crying in the recovery and is inconsolable, how would you manage the situation?
Mini-CEX	GA/IPPV case, ASA 2-3, age 3-5 years	1. Management of anticipated/ unanticipated difficult airway (mask ventilation/ intubation) 2. Strategies for nasal intubation 3. Extubation strategy 4. PONV 5. Fluid management	1. What are the implications of OSA in young children? If this patient had a STUBR score of 3, how would that determine your management of the patient 2. If your patient had to have nasal intubation and you encountered difficulty advancing the tracheal tube down below the cords, what would you do to circumvent the problem? 3. How do you risk stratify for PONV in children and decide on the appropriate PONV prophylaxis for our current patient? 4. What happens if your patient develops stridor post extubation? 5. How are you going to manage fluids in this patient and why?
EBD	Paediatric trauma patient requiring surgery	1. Assessment of paediatric trauma patient 2. Management of paediatric trauma patient a. Modified RSI b. Fluid management c. Airway management, emergency d. Drug doses e. Massive haemorrhage 3. Paediatric APLS- discussion regarding resuscitation	1. How would you assess a paediatric trauma patient and what in particular would you ask in the history from the caregivers? 2. What is modified rapid sequence induction and how would you perform that? 3. If this patient were a 5-year-old who had been hit by a car and arrives at ED with a GCS of E3V3M3, HR of 160 and BP of 70/40mmHg with a CT scan that shows an extradural hematoma a. How would you conduct the anaesthetic for this patient?

		4. Differences in paediatric trauma vs adult trauma	b. If there is difficulty getting another peripheral venous line for this patient, how would you proceed? c. What would your anaesthetic goals be?
EBD	Special needs child requiring anaesthesia	1. Preoperative assessment and preparation of special needs child 2. Role of premedication 3. Role of care giver in perioperative management 4. Management of lines and tubes postoperatively 5. Prevention of iatrogenic injury	1. If you saw this child in the pre-anaesthetic clinic, how would you counsel the parents and what can you do to better prepare the child for surgery and make the perioperative experience a better one for parent and child? 2. If we were to give premedication to this child, what would your choice of drugs be and the route? Are there maximum doses you need to be mindful of? 3. What is our emergence strategy going to be for this child and why do you say that?

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<table>
<tr><td>Title</td><td>Managing adult patients for cardiovascular procedures and interventions</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td>Specifications:
1. Performing risk assessment and optimising medical comorbidities prior to interventions
2. Preparing equipment and medications for cardiovascular procedures and interventions with attention to maintaining sterility of invasive vascular lines
3. Communicating care plans with anaesthesia and interprofessional teams.
4. Providing perioperative care of patients for cardiovascular procedures and interventions.
5. Providing sedation and intravenous anaesthetics for interventional procedures.
6. Managing crises
7. Transferring patients to post-anaesthesia care unit or intensive care unit.
8. Conducting handover of patient care.</td></tr>
<tr><td>Limitations:
This EPA excludes the following:
1. Patients aged less than 16 years old</td></tr>
<tr><td>Sub competencies relevant for this EPA</td><td>
<table>
<tr><td>PC1</td><td>✓</td><td>PC7</td><td>✓</td><td>SBP1</td><td></td><td>P1</td><td>✓</td></tr>
<tr><td>PC2</td><td>✓</td><td>PC8</td><td>✓</td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
<tr><td>PC3</td><td>✓</td><td>PC9</td><td></td><td>SBP3</td><td></td><td>P3</td><td></td></tr>
<tr><td>PC4</td><td>✓</td><td>PC10</td><td></td><td>PBL 1</td><td>✓</td><td>ICS1</td><td>✓</td></tr>
<tr><td>PC5</td><td>✓</td><td>MK1</td><td>✓</td><td>PBL 2</td><td>✓</td><td>ICS2</td><td>✓</td></tr>
<tr><td>PC6</td><td>✓</td><td>MK 2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
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</td></tr>
</table>

Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	**1. Basic science** • Anatomy and physiology of the cardiovascular system • Applied anatomy of vascular access for invasive monitoring • Guideline-directed medical therapy for cardiovascular diseases. • Pharmacology of medications used during cardiovascular procedures and interventions. • Equipment (anaesthetic machine, monitoring, and airway equipment) **2. Investigative tools** • Coronary angiogram • Sestamibi (MIBI) scan • Stress tests (pharmacologic, exercise) **3. Monitoring tools** • Transoesophageal echocardiography • Monitoring equipment (Interpreting waveforms and parameters on different types of hemodynamic monitoring devices e.g., equipment for measuring cardiac output, pulmonary artery catheter) • Cerebral monitoring (e.g., cerebral oximetry monitoring, processed EEG monitoring) • Coagulation profile (activated clotting time, thromboelastography, rotational thromboelastometry) **4. Cardiopulmonary support** • Mechanical ventilation (Indications, modes of ventilation, complications, maintenance, lung recruitment manoeuvres, weaning and extubation criteria.) • Humidification (indications, modes, complications) • Pacemaker (temporary and permanent; implantable cardioverter defibrillator) • Extracorporeal circuit (cardiopulmonary bypass circuit, veno-venous and venoarterial extracorporeal membrane oxygenator circuits) • Types of mechanical circulatory support **5. Organ protection** • Cardioprotection • Neuroprotection (cerebrovascular accident, postoperative delirium, postoperative cognitive dysfunction) • Spinal cord protection in repair of descending aortic aneurysm • Renal protection • Thermoregulation (indications, modes, complications) **6. Cardiovascular procedures and interventions** • Coronary artery grafting surgery (on pump, off-pump) • Open-heart procedures • Transcatheter aortic valve implantation • Radiofrequency ablation of atrial fibrillation procedure • Percutaneous closure of atrial septal defect

		• Surgical repair of aorta (ascending, arch of, and descending) **7. Institution-specific processes** • Timeout, sign-in, and sign-out processes. • Institution-specific massive blood transfusion protocol • Antibiotic prophylaxis **8. Cardiovascular diseases or disorders** • Hypertensive disease • Diastolic dysfunction • Ischaemic heart disease • Stable chronic heart failure • Cardiac arrhythmias • Types of aortic aneurysm • Aortic dissection • Types of pulmonary hypertension • Pulmonary thromboembolism • Vasoplegia • Low cardiac output syndrome **9. Gastroenterology** • Gastric oesophageal reflux disorder **10. Haematology** • Anaemia • Thrombocytopenia • Coagulation disorders • Systemic inflammatory response • ABO-incompatibility **11. Nephrology** • Cardiac surgery associated acute kidney injury • Chronic renal failure • Electrolyte disturbances • Discoloured urine **12. Neurology** • Seizures • Cerebrovascular disease (e.g., amaurosis fugax, transient ischaemic attack, stroke) **13. Infection control** • Central venous catheter associated infection • Deep sternal wound infection

	S:	1. Clinical assessment skills • History taking/ physical examination/ ordering and interpretation of relevant investigations 2. Procedural skills • Management of airway • Intravascular access (central venous/ arterial) • Ultrasonography skills 3. Communication and interpersonal skills • Patient-centric communication including counselling and taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including team brief/huddling and accurate documentation 4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management, time management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to supervisor when appropriate, 4. Provides efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Collegial and respectful team player who is willing to help others
	E:	1. Completion of 12 weeks of cardiothoracic anaesthesia module. 2. E-learning modalities 3. Transoesophageal echocardiography simulation 4. Case Logs
Potential risks in case of failure	1. Increase in patient morbidity/ dissatisfaction/mortality. 2. Increase in medico-legal liability. 3. Inappropriate over-utilisation of healthcare resources/ increase healthcare cost	

Sources of information to support summative entrustment decisions	**Tools**	**Number to be completed satisfactorily**	**Additional specifications if needed (To consider evaluators and context)**
	Direct Observation of Procedural Skills (DOPS)	Minimum 3 by end R2 reaching minimal score of 5 for overall assessment.	Physician Faculty *To be completed by end of R2* 1. Central venous catheter insertion
	Entrustment-based Discussion (EBD)	Minimum 1 by end R3 For level 3 supervision	Physician Faculty co-managing case *To be completed by end of R3* 1. Patient undergoing cardiac surgery requiring CPB

	Competency Evaluation	2	Physician Faculty • 1 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
	Multisource 360° Feedback (MSF)	2	Minimum 1 physician faculty, 1 peer and 1 nurse (Staff Nurse and above) • 1 in Junior Residency (R1 to R3) • 1 in Senior Residency (R4 to R5)
	Case Logs	Minimum case numbers 20	Reviewed by programme *To be completed by end of R3* At least 20 patients for cardiac surgery and cardiology intervention • At least 10 procedures requiring cardiopulmonary bypass. • At least 10 procedures not requiring cardiopulmonary bypass, e.g., insertion of implantable cardioverter defibrillator or pacemaker, cardiac ablation, repair of descending aortic aneurysm, and pericardiotomy surgery.
		Minimum case numbers 15	*To be completed by end of R5* 15 patients for cardiac surgery and cardiology interventions • At least 5 procedures must involve cardiopulmonary bypass
Which supervision level when?	Level 3 supervision at end R5 Resident is expected to exit the programme at level 3 supervision because cardiac anaesthesia is a subspecialty		
Expiry	If not practiced, after how many months or years will more close supervision be required again?		1 year

Entrustment Questions for EPA 10

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
DOPS	Central Venous catheter (CVC)	1. Appropriate consent taking- explains indication, risks, benefits 2. Understands indications and contraindication to CVC placement 3. Communicates with patient during procedure to ensure comfort, uses LA 4. Appropriate monitoring during insertion including ECG 5. Appropriate checking of equipment, including ultrasound machine and transducer 6. Appropriate aseptic technique 7. Knows landmark technique (accurately locates anatomical landmarks) in addition to using ultrasound guided placement 8. Proper technique 9. Checks the guidewire position on ultrasound after insertion 10. Ensures that direction for dilation of track does not result in bent guidewire 11. Communicates needs clearly with healthcare team members 12. Escalates to senior appropriately when difficulty arises 13. Demonstrates situational awareness during procedure 14. Manages sharps appropriately and safely disposes of them 15. Appropriate securing of CVC 16. Documents procedure properly 17. Orders and checks chest X ray post CVC insertion	1. Describe the anatomy, surface marking, landmarks for internal jugular venous and subclavian venous routes of CVC insertion. 2. Identify the structures in the ultrasound scan you have acquired. 3. What are the complications associated with insertion of CVC? 4. What if the patient developed atrial arrhythmias after insertion of the guidewire? 5. After the drapes are removed, the patient is noted to be extremely pale, the heart rate has increased from 80/min to 95/min and the blood pressure has decreased from 160/90mmHg to 100/78mmHg. What do you think has happened and what would you do next? 6. What would you look out for on the chest X ray after CVC insertion.
EBD	Patient undergoing cardiac surgery requiring CPB	1. Discuss methods of cardiology investigations and evaluations, and interpretation and perioperative application of findings of these investigations. a. treadmill exercise	1. What are the objectives of the treadmill exercise and how can you apply the findings to clinical practice. 2. What is the MIBI scan and how is it done? What implications are there for risk stratification and CABG?

		b. Dipyridamole - ^{99m}Technetium sestaMIBI (methoxybutyl isonitrile) scan c. cardiac CT calcium score aka coronary calcium scan d. coronary angiogram 2. Discuss CABG a. How to manage preoperative and intraoperative period up to commencement of bypass. b. How to manage a haemodynamically unstable patient c. Indications for cardiopulmonary bypass (CPB) or off CPB CABG d. Pharmacokinetics of heparin/ protamine e. What is heparin resistance and heparin induced thrombocytopenia and how do you overcome these conditions? 3. Interpretation of echocardiography images a. Evaluation of volume status and ventricular contractility. b. Detection of air in the heart chambers. 4. The principles of pacing the heart. a. Pacing modalities and codes b. What mode and at what rate that makes sense for patients who develop new onset dysrhythmia in non-cardiac surgery settings c. Precautions necessary for a patient with pacemaker/cardiac implantable electronic device undergoing non-cardiac surgery 5. The principles and interpretation of thromboelastography (TEG) and rotational thromboelastometry (ROTEM)	3. If the patient has had cardiac calcium CT score aka coronary calcium scan, what are the implications of the findings? 4. In this patient's coronary angiogram, can you identify the main coronary arteries and sites of blockage? What implications are there for us? 5. Can you explain critical stenosis of an artery? How does a critical stenosis affect the flow in the distal vascular bed at rest? 6. When would interventional procedure be considered for a patient with coronary artery disease? 7. If acute ST depression is noted on the ECG during surgery, how would you manage the patient? 8. How would you manage the patient's haemodynamic responses just before aortic cannulation and during CPB for this patient undergoing CABG? 9. What organ protective strategies can be employed in this patient during CPB 10. What conditions are required to wean a patient off CPB? 11. Which patients would benefit from a trans-oesophageal echocardiography in a non-cardiac setting. Would you be able to demonstrate the different views in this patient and tell me what you see? 12. What precautions are necessary for a patient with pacemaker/cardiac implantable electronic device undergoing non-cardiac surgery? 13. Interpret this ROTEM/TEG printout and describe your management? 14. How do you recognise cardiac tamponade? 15. After CABG, this patient becomes acutely unstable while being transferred from the OT table to the bed. What would your management be? 16. How would you manage patients for pericardiocentesis (drainage of pericardial effusion to relieve tamponade)?

		6. Management of surgical drainage of pericardial effusion a. Describe the pathophysiology of cardiac tamponade. b. Describe the investigations. c. Describe the perioperative preparation and management for pericardiocentesis.	

Anaesthesiology EPA 11

<table>
<tr><td>Title</td><td colspan="2">Assessing and managing patients with acute pain</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td colspan="2">Specifications:
1. Performing assessment using focused history, targeted physical examination and selecting appropriate investigations to gather relevant information
2. Diagnosing common causes of acute pain based on gathered information
3. Developing and implementing patient-centric pain management strategies using a multimodal approach, appropriate equipment and techniques based on best available evidence/ guidelines
4. Discussing with patient +/- family on pain management options and their potential side effects, and setting pain treatment goals together
5. Documenting all clinical encounters including consent taking for procedures, risk benefit discussions, agreed pain management plan
6. Coordinating care and communicating with all healthcare members using verbal, written or electronic medical record, or other digital technology
7. Monitoring for and managing the complications of various acute pain management strategies
8. Recognising the need to seek senior or specialist opinion/help when appropriate</td></tr>
<tr><td colspan="2">Limitations:
This EPA excludes the following:
1. Pure chronic pain cases
2. Patients aged less than 16 years old</td></tr>
<tr><td>Sub competencies relevant for this EPA</td><td colspan="2">
<table>
<tr><td>PC1</td><td></td><td>PC7</td><td></td><td>SBP1</td><td></td><td>P1</td><td></td></tr>
<tr><td>PC2</td><td>✓</td><td>PC8</td><td></td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
<tr><td>PC3</td><td></td><td>PC9</td><td></td><td>SBP3</td><td></td><td>P3</td><td></td></tr>
<tr><td>PC4</td><td></td><td>PC10</td><td>✓</td><td>PBL 1</td><td>✓</td><td>ICS1</td><td>✓</td></tr>
<tr><td>PC5</td><td></td><td>MK1</td><td>✓</td><td>PBL 2</td><td></td><td>ICS2</td><td>✓</td></tr>
<tr><td>PC6</td><td></td><td>MK 2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
</table>
</td></tr>
<tr><td rowspan="2">Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment</td><td>K:</td><td>1. Applied anatomy and physiology of acute pain
2. Assessment of pain
3. Pharmacology of analgesics and adjuvants
4. Role of pain management in surgical outcome/ patient recovery
5. Multimodal pain management strategies including non-pharmacological options
6. Equipment for analgesia including catheters, pumps
7. Management of complications related to pain management
8. National guidelines on safe prescription of opioids
9. Organisation and requirements of an institution based acute pain service</td></tr>
<tr><td>S:</td><td>1. Clinical assessment skills</td></tr>
</table>

<table>
<tr><td rowspan="4"></td><td></td><td colspan="3">• History taking/ physical examination/ ordering and interpretation of relevant investigations in relation to pain
2. Procedural skills
• Regional anaesthesia blocks
3. Communication and interpersonal skills
• Patient-centric communication including counselling, taking informed consent
• Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including accurate documentation while maintaining patient confidentiality
4. Non-technical skills (e.g., prioritisation, situation awareness, ability to organise, critical decision making, leadership, crisis resource management</td></tr>
<tr><td>A:</td><td colspan="3">1. Proactive in reflective learning and self- improvement, receptive to feedback
2. Honest, reliable, conscientious, responsible
3. Recognises own limits and escalates to supervisor when appropriate,
4. Provides efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality
5. Collegial and respectful team player who is willing to help others</td></tr>
<tr><td>E:</td><td colspan="3">1. Acute Pain service (minimum 1 week/year)
2. Post Anaesthetic Care Unit during junior residency (minimum 2 weeks/year R1 or R2)
3. Teaching sessions on pain management
4. Case log reflections</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Potential risks in case of failure</td><td colspan="4">1. Increase in patient dissatisfaction/morbidity with potential for mortality e.g., opioid overdose
2. Increase in medico-legal liability
3. Increase in healthcare cost</td></tr>
<tr><td rowspan="5">Sources of information to support summative entrustment decisions</td><td colspan="2">Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed (To consider evaluators and context)</td></tr>
<tr><td colspan="2">Mini-Clinical Examination (Mini-CEX)</td><td>Minimum 1 by end R3 for level 3 supervision</td><td>Physician Faculty
To be completed by end R3
1. To cover pain assessment and acute pain management (PACU/ pain rounds)</td></tr>
<tr><td colspan="2">Entrustment-based Discussion (EBD)</td><td>Minimum 1 by end R5 for level 4 supervision</td><td>Physician Faculty co-managing case
To be completed by end R5
1. Blue letter referral setting</td></tr>
<tr><td colspan="2">Case Logs</td><td>Minimum 20</td><td>Reviewed by programme
To be completed by end R3
• Evaluation of new patient presenting with pain</td></tr>
<tr><td colspan="2">Case reflections</td><td>Minimum 5</td><td>To be completed by end R5
Challenging cases of acute pain management</td></tr>
</table>

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Which supervision level when?	Level 3 supervision by end R3 Level 4 supervision by end R5	
Expiry	If not practiced, after how many months or years will more close supervision be required again?	1 year

Entrustment Questions for EPA 11

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Pain assessment and acute pain management (PACU/Pain rounds)	1. Assessment of the patient's pain via history taking/ physical examination and investigations as appropriate 2. Management of acute pain	1. What did you gather after speaking to the patient/family/carer? 2. What findings did you obtain with clinical examination? 3. What investigations did you look at or what further investigations would you like to have? 4. What is your clinical impression/main issues in this case? 5. What would be the differential diagnosis? 6. How would you manage this case and why did you choose this management strategy? 7. What are the considerations or your concerns in the management of this case? 8. Who have you involved or who do you need to involve in your management/communications? 9. How would you proceed if patient or family do not agree with your plan? 10. What would you do if patient cannot tolerate the medication you prescribed? 11. What would you do if your investigations showed critical abnormal findings e.g., new spinal stenosis/disc bulge on MRI? 12. How would you change your management if the patient has renal dysfunction or on anticoagulants/dual anti-platelets? 13. How do feel about your management for this case? 14. How else could you have managed this case differently/better? 15. Do you feel your experience and knowledge are adequate to manage this case independently? If not, how would you prepare/equip yourself better?

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EBD	Management of Blue letter referral for acute pain	1. Review by faculty of documentation of blue letter reply by resident with feedback provided by faculty on the quality of reply 2. Communication with primary team and other members of the healthcare team such as nurses and allied health	1. How did you approach this case? 2. What did you gather from the history/physical examination? 3. Would you ask for any investigations and why? 4. What is your clinical impression/what are the main issues in this case? 5. What are the considerations for your management strategy? 6. What would your management strategy be? 7. What if the primary team/ patient/ family disagrees with your management strategy? 8. What if the patient is non-verbal but groans constantly? 9. How should this patient be monitored and followed up? 10. Are there strategies you know of to prevent the development of chronic pain

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<table>
<tr><td>Title</td><td>Assessing and managing patients with chronic pain</td></tr>
<tr><td rowspan="2">Specifications and limitations</td><td>Specifications:
1. Performing assessment with focused history-taking, physical examination, and relevant investigations to ascertain the following:
i. Biological basis for pain symptoms
ii. Psychological state of the patient
iii. Social impact of pain such as ability to function, rest and pursue interests and goals
2. Diagnosing common chronic pain conditions such as spinal pain, neuropathy
3. Formulating patient-centric, multimodal, multidisciplinary management chronic pain management plans using the biopsychosocial model approach
4. Consulting pain specialist or allied health care professionals appropriately
5. Communicating with patients and their families to allay fears and manage expectations, make informed decisions regarding their chronic pain condition, and counsel regarding treatment strategies including potential side effects of medications or complications of pain treatment procedures
6. Coordinating care and communicating with all healthcare members using verbal, written or electronic medical record, or other digital technology</td></tr>
<tr><td>Limitations:
This EPA excludes the following:
1. Patients aged less than 16 years old</td></tr>
<tr><td>Sub competencies relevant for this EPA</td><td>
<table>
<tr><td>PC1</td><td></td><td>PC7</td><td></td><td>SBP1</td><td></td><td>P1</td><td>✓</td></tr>
<tr><td>PC2</td><td>✓</td><td>PC8</td><td></td><td>SBP2</td><td>✓</td><td>P2</td><td>✓</td></tr>
<tr><td>PC3</td><td></td><td>PC9</td><td></td><td>SBP3</td><td></td><td>P3</td><td></td></tr>
<tr><td>PC4</td><td></td><td>PC10</td><td>✓</td><td>PBL 1</td><td>✓</td><td>ICS1</td><td>✓</td></tr>
<tr><td>PC5</td><td></td><td>MK1</td><td>✓</td><td>PBL 2</td><td></td><td>ICS2</td><td>✓</td></tr>
<tr><td>PC6</td><td></td><td>MK 2</td><td>✓</td><td></td><td></td><td>ICS3</td><td>✓</td></tr>
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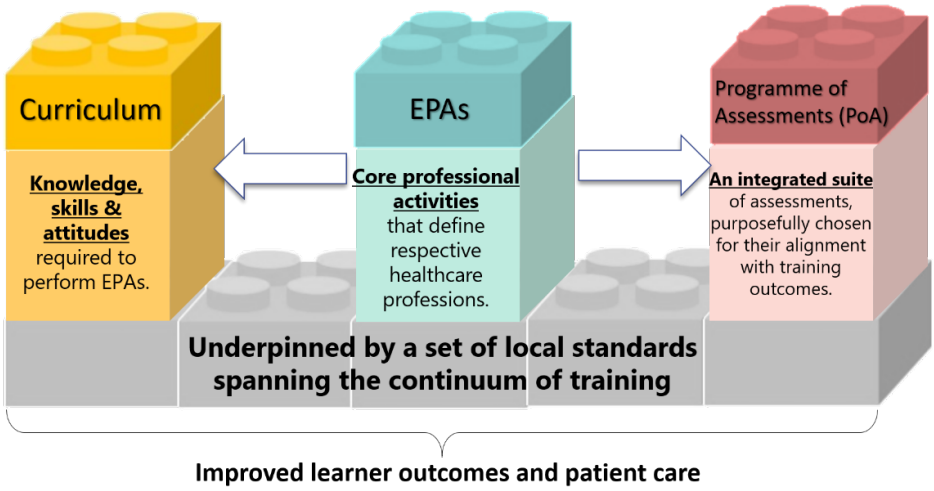
Required Knowledge, Skills, Attitudes and Experiences, to enable summative entrustment	K:	**1. Multimodal and multidisciplinary management plan of**: a. Nociceptive pain: b. Neuropathic pain c. Inflammatory conditions d. Central sensitisation conditions **2. Basis of, indications and contra-indications of, effects of and evidence for**: a. Pharmacological therapy of neuropathic pain b. Pharmacological therapy of chronic visceral and somatic pain c. Opioid usage in chronic pain d. Interventional pain therapy e.g., Facet joint injection **3. Knowledge in clinical sciences relevant to pain medicine** **4. Multimodal and multidisciplinary management plan for situations**: a. Opioid tolerant, opioid dependent or addicted patients presenting with acute pain b. Altered physiology: Extremes of age (young and old), pregnancy and breastfeeding, c. Co-morbidities such as obstructive sleep apnoea, renal or hepatic impairment. d. Patients with cancer **5. Biopsychosocial model of pain management** **6. Role of pharmacogenetics in pain management** **7. Role of non-pharmacological therapy such as psychology/ mindfulness**
	S:	1. Clinical assessment skills • Chronic pain history taking including character, severity, impact of pain on function/ physical examination/ ordering and interpretation of relevant investigations in relation to pain such as MRIs, X –rays, blood investigations (e.g., inflammatory markers, HLA genotyping etc.) 2. Procedural skills • Regional anaesthesia blocks 3. Communication and interpersonal skills • Patient-centric communication including counselling, taking informed consent • Healthcare professional/ interdisciplinary centric communication to coordinate multidisciplinary care of patient including accurate documentation while maintaining patient confidentiality 4. Non-technical skills (e.g., resource management)
	A:	1. Proactive in reflective learning and self- improvement, receptive to feedback 2. Honest, reliable, conscientious, responsible 3. Recognises own limits and escalates to experts when appropriate, 4. Provides efficient patient-centric compassionate care in a timely manner while ensuring patient comfort, privacy, and confidentiality 5. Behaves professionally even under stressful or complex situations 6. Collegial and respectful team player who is willing to help others

<table>
<tr><td></td><td colspan="3">E:
1. Chronic pain services including clinics and pain intervention sessions
a. minimum of 2 weeks in R1/R2
b. minimum of 2 weeks in R3/ Senior Residency (R4 to R5)
2. Teaching sessions on chronic pain management</td></tr>
<tr><td>Potential risks in case of failure</td><td colspan="3">1. Increase in patient dissatisfaction/morbidity including depression
2. Increase medico-legal liability
3. Increase healthcare cost
4. Delay in diagnosis and treatment</td></tr>
<tr><td rowspan="4">Sources of information to support summative entrustment decisions</td><td>Tools</td><td>Number to be completed satisfactorily</td><td>Additional specifications if needed
(To consider evaluators and context)</td></tr>
<tr><td>Mini-Clinical Examination (Mini-CEX)</td><td>Minimum 1 by end R3 for level 3 supervision</td><td>Physician Faculty
To be completed by end R3
1. Assessing chronic pain
• History and physical examination</td></tr>
<tr><td>Entrustment-based Discussion (EBD)</td><td>Minimum 1 by end R3 for level 3 supervision</td><td>Physician Faculty co-managing case
To be completed by end R3
1. Complex chronic pain patient coming in for surgery
• Pain management for opioid tolerant patient</td></tr>
<tr><td>Multisource 360° Feedback (MSF)</td><td>1</td><td>Pain nurse/ physician faculty running chronic pain clinic
To be completed by end R3</td></tr>
<tr><td>Which supervision level when?</td><td colspan="3">Level 3 supervision by end R5
Resident is expected to exit the programme at level 3 supervision because chronic pain management is a subspecialty</td></tr>
<tr><td>Expiry</td><td colspan="2">If not practiced, after how many months or years will more close supervision be required again?</td><td>1 year</td></tr>
</table>

Entrustment Questions for EPA 12

Assessment tool	Specifications	Discussion Points / Expectations	Suggested entrustment questions
Mini-CEX	Assessing chronic pain	History and physical examination	1. What did you gather after speaking to the patient/family/carer? 2. What findings did you obtain with clinical examination? 3. What investigations did you look at or what further investigations would you like to have? 4. What is your clinical impression/main issues in this case? 5. What would be the differential diagnosis? 6. How would you manage this case? What would your management strategy be? 7. Why did you choose the management you mentioned? 8. What would you do if your investigations showed critical abnormal findings e.g., new spinal stenosis/disc bulge on MRI? 9. What are the considerations or your concerns in the management of this case? 10. What would prompt you to refer to a chronic pain specialist? 11. What kind of interventions are there for chronic pain that you know of?
EBD	Chronic pain patient coming in for surgery	Pain management for opioid tolerant	1. What is your clinical impression/main issues in this case? 2. What are the considerations or your concerns in the management of this case? 3. How would you manage this case/What would your management strategy be? 4. Why did you choose the management you mentioned? 5. What problems do you anticipate with the patient being on long term opioids or medical cannabis?

Appendix



How are WBAs linked to EPAs?

EPAs operationalise competencies into actual practice by placing focus on the performance of actual tasks. WBAs allow a direct assessment of the performance of these tasks, and in turn allow (i) identification of resident competency gaps; and (ii) a decision to be made on residents' level of readiness to perform the task. Entrustment decisions about EPAs require longitudinal mentorship, and having sufficient supervision and feedback is key to good entrustment decisions. (ten Cate Olle, Hart and Ankel, 2016)

WBAs are formative and provide evidence to support entrustment of

- **To support entrustment decisions on EPAs**

With WBAs, individual supervisors can make a decision on the level of entrustment to perform the task at that point in time ('formative decision') or during the period where the resident is under their supervision. For a summative entrustment decision to be made, the Clinical Competency Committee (CCC) should review multiple WBAs completed by multiple assessors at multiple timepoints, to gain a confident and holistic view of that the resident is able to prospectively perform at that level of entrustment. WBAs, therefore, provide critical sources of information on the resident's competency level to support and inform entrustment decisions. The minimum number of WBAs that must be reviewed will be included in the respective EPA templates as Sources of Information.

Common types of WBAs (taken from the 'Ins and Outs of Entrustable Professional Activities' an international course conducted by faculty)

Approaches	Methods	Tools
Watching	Brief and focused observation	Mini-Clinical Examination (Mini-CEX) Direct Observation of Procedural Skills (DOPS) Simulation (Patterson et al., 2013) Audio-based evaluation (Sanatani et al., 2020)
	Longitudinal observation	Multisource 360° Feedback (MSF)
Talking	Brief conversations	Case-based discussions Chart-Stimulated Recall One-minute Preceptor SNAPPS Entrustment-based discussions
Reviewing results	Product evaluation	Entries in Electronic Health Record; Discharge letters; QI reports; Resident-Sensitive Quality Measures (RSQM) (Schumacher et al., 2018)